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WATERMARKABLE MULTI-DRAFT SPECULATIVE SAMPLING VIA POISSON PROCESSES

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ABSTRACT

Large language models (LLMs) have achieved state-of-the-art performance across a wide range of tasks, motivating two important aspects of deployment: inference efficiency and output provenance, which can be tackled by speculative sampling and watermarking, respectively. However, recent works have shown that combining these two goals is highly nontrivial and can be potentially impossible. In this work, we develop a novel multi-draft speculative sampling algorithm based on Poisson processes that improves the frontier of this fundamental trade-off. The proposed algorithm has strong sampling efficiency on its own and, more interestingly, is naturally *watermarkable*: we can embed an unbiased watermark without degrading speculative acceptance. Moreover, our algorithm is based on an *exact* list-coupling-without-communication scheme, which yields a *drafter invariance* property that benefits both sampling and watermarking. It is the first multi-draft, drafter-invariant speculative sampling scheme that maintains both watermark strength and sampling efficiency, and we experimentally verify its strong performance in both aspects.

1 INTRODUCTION

Large language models (LLMs) have become a central component of modern generative AI, achieving strong performance in dialogue, reasoning, code generation, and open-ended content creation. Their probabilistic nature based on autoregressive sampling enables them to generate diverse responses (Brown et al., 2020; Grattafiori et al., 2024; Singh et al., 2025). However, as LLM-generated text becomes increasingly fluent and difficult to distinguish from human-written content, concerns about accountability, provenance, misuse, and intellectual property have attracted increasing attention (Wu et al., 2025). For instance, ChatGPT-generated scientific abstracts are hard for human reviewers to distinguish from human-generated ones, yet they may contain fabricated data (Shumailov et al., 2024). Hence, watermarking has become an important, and potentially necessary, tool for tracing the provenance of LLM-generated text by embedding statistical signals into the sampling process while preserving output quality (Kirchenbauer et al., 2023; Aaronson and Kirchner, 2023; Kuditipudi et al., 2023). For instance, reflecting its growing importance, Anthropic recently announced that future Claude models will generate text with embedded watermarks (Anthropic, 2026).

Apart from provenance, the efficiency of autoregressive generation is another bottleneck in the system pipeline. *Speculative sampling* (also known as decoding), has emerged as a promising approach to accelerate content generation without changing the output distribution of the target model (Chen et al., 2023; Leviathan et al., 2023). However, speculative sampling is not independent of watermarking. On the contrary, recent works (Hu and Huang, 2024; He et al., 2026) show that combining the two is highly nontrivial: in (Hu and Huang, 2024), a "no-go" theorem was proved, showing that one cannot embed a watermark without degrading the performance of speculative sampling, thereby establishing a fundamental trade-off between watermark strength and speculative sampling efficiency. He et al. (2026) recently improves this trade-off by considering alternative watermark metrics.

In this work, we ask the following question: can we design a watermarkable speculative sampling scheme that further improves the trade-off (Hu and Huang, 2024; He et al., 2026) by increasing speculative sampling efficiency without compromising watermark detectability?

We answer this question in the **affirmative**: we design a novel multi-draft speculative sampling algorithm that achieves strong acceleration performance, can naturally embed unbiased watermark, and, more importantly, has watermark strength that does not degrade as the sampling efficiency

increases.[1] Our multi-draft construction relies on Poisson processes (Maddison, 2016; Li and Anantharam, 2021) and yields a *list-coupling without communication* scheme, which endows our algorithm with a *drafter-invariance* property (Daliri et al., 2025; Rowan et al., 2025) that becomes more desirable and essential in the context of watermarking. Our contributions are two-fold:

- **Multi-draft speculative sampling**: We design a novel multi-draft speculative sampling algorithm that generates and verifies multiple draft continuations. Our scheme uses desirable properties of Poisson processes, which naturally yield a drafter invariance property (Daliri et al., 2025). Compared with (Rowan et al., 2025), which has been shown to outperform other existing methods, we theoretically prove that our acceptance rates are higher and empirically demonstrate better acceleration performance across multiple models and datasets.
- **Efficiency-preserving watermarking**: Besides desirable sampling efficiency, our algorithm is naturally *watermarkable* with the use of Poisson processes, which can embed a recoverable statistical signal while maintaining the distribution *unbiased*. By building the watermark directly into the sampling procedure through keyed randomness, we maintain watermark strength while preserving sampling efficiency, thereby improving the trade-off frontier identified in (Hu and Huang, 2024) and achieving better overall performance.

2 Related Works

LLM Decoding and Speculative Sampling. Standard autoregressive decoding methods, including top-k sampling (Radford et al., 2019; Fan et al., 2018), nucleus sampling (Holtzman et al., 2020), and the permute-and-flip decoder (Zhao et al., 2025a), generate tokens sequentially; and therefore, suffer from the inherent latency of one target-model call per token (Stern et al., 2018). Speculative sampling (Chen et al., 2023; Leviathan et al., 2023) addresses this bottleneck by using a smaller draft model to propose several future tokens, while the larger target model verifies the proposed continuations in parallel. The efficiency gain depends on how well the draft distribution matches the target distribution. Recent works further improve this paradigm through optimal-transport-based token selection (Sun et al., 2023), tree-based inference and verification (Miao et al., 2024), and multi-draft architectures (Khisti et al., 2025; Rowan et al., 2025). It can also be used to accelerate diffusion models (De Bortoli et al., 2025; Bullo et al., 2026).

Watermarking. LLM watermarking aims to embed detectable statistical signals into generated text while preserving fluency and utility. Early logit-bias methods pseudorandomly select green-list tokens and detect the resulting bias by hypothesis testing (Kirchenbauer et al., 2023), while later works improve robustness, accessibility, and probability balance (Wu et al., 2023; Giboulot and Furon, 2024; Park et al., 2026). A parallel line studies unbiased watermarking, where the sampler maintains the distribution while correlating the output with secret randomness (Kuditipudi et al., 2023; Hu et al., 2023; Christ et al., 2024). They connect watermarking to information theory (Moulin and O'Sullivan, 2003) and statistics, including detection power, false-positive control, optimal tests, and couplings (Li et al., 2025; He et al., 2025; Tsur et al., 2025). Related works include robust or multi-bit attribution (Zhao et al., 2023; Boroujeny et al., 2024; Qu et al., 2025), and efficient identification of watermarked segments (Zhao et al., 2025b). In recent works (Hu and Huang, 2024; He et al., 2026) an inevitable trade-off between speculative sampling and watermarking has been discussed.

Poisson Functional Representation (PFR). PFR is a sampling scheme introduced in (Li and El Gamal, 2018), with related Poisson process constructions also discussed by Maddison et al. (2014); Maddison (2016). It selects a sample from a target distribution through a Poisson process and a proposal distribution, and has been applied to information theory settings (Li and Anantharam, 2021; Khisti et al., 2024; Liu and Li, 2024; 2025; Liu et al., 2026) and machine learning (He et al., 2024; Liu et al., 2024; Zhou et al., 2026; Flamich et al., 2026). It can be extended to list-coupling (Li and Anantharam, 2021) by the mapping theorem (Last and Penrose, 2017) of Poisson process, and achieves the best known bound on coupling without communication (Daliri et al., 2025).

Notations. $x_{i:j}$ denotes the sequence $x_i, x_{i+1}, \ldots, x_j$, and we let $x_{:j} := x_{1:j}$. $P \ll Q$ denotes that probability measure P is absolutely continuous with respect to Q. The logarithm is on base 2.

[1] Our multi-draft scheme does not contradict the inevitable trade-off (Hu and Huang, 2024): it still applies for a fixed number of drafts. We improve the trade-off along an additional dimension, namely, the number of drafts.

3 SPECULATIVE SAMPLING

We first briefly review the technical background of speculative sampling (Chen et al., 2023; Leviathan et al., 2023; Sun et al., 2023; Rowan et al., 2025). Given volcabulary $\mathcal{V}$ and a context $x_{:t} := (x_1, x_2, \ldots, x_t)$, an autoregressive language model $\mathcal{M}_b$ generates the next token by sampling from the conditional distribution $\mathcal{M}_b(\cdot|x_{:t})$ under temperature sampling (Ackley et al., 1985; Ficler and Goldberg, 2017). *Speculative sampling* accelerates this process by introducing a smaller and cheaper *draft model* $\mathcal{M}_s$, which proposes several candidate tokens before the target model verifies them. Given a prefix $x_{:t}$, one iteration of the speculative sampling consists of the following three stages:

1. **Draft Stage.** The draft model $\mathcal{M}_s$ sequentially samples L candidate tokens $\tilde{x}_{t+1}, \ldots, \tilde{x}_{t+L}$. For each position $i = 1, \ldots, L$, we record the draft distribution $\mathcal{M}_s(\cdot|x_{:t}, \tilde{x}_{t+1:t+i-1})$.
2. **Computation Stage.** Conditioned on the drafted context, the target model $\mathcal{M}_b$ evaluates the corresponding conditional distributions, potentially in parallel:
$$\mathcal{M}_b(\cdot|x_{:t}), \mathcal{M}_b(\cdot|x_{:t}, \tilde{x}_{t+1}), \ldots, \mathcal{M}_b(\cdot|x_{:t}, \tilde{x}_{t+1:t+L}),$$
3. **Selection Stage.** By the distributions $\mathcal{M}_s$ and $\mathcal{M}_b$, we accept the longest prefix of the drafted tokens until the first rejection, and a *correction token* is sampled from a specific residual distribution. If all L drafts are accepted, one more token from the target is sampled.

The acceptance-correction procedure ensures that the generated tokens follow *exactly* the same distribution as direct autoregressive sampling, i.e., improving decoding efficiency does not change the target law. The exactness is achieved via a recursive token-level maximal coupling (Algorithm 1), where we use P, Q to denote the target and draft distributions, respectively, and achieve

$$\Pr(X = Y) = \sum\nolimits_{x \in \mathcal{V}} \min(P(x), Q(x)) = 1 - \|P - Q\|_{\mathrm{TV}},$$

i.e., the closer P and Q are, the higher the probability that the draft token will be accepted. Together with the extra sampled token, there could be at most $L + 1$ tokens generated in each iteration, and therefore, the speedup is up to $(L + 1)$ times if the decoding time of the draft model is negligible.

Kobus and Gündüz (2025) proposed a new sampling scheme (Algorithm 2) based on exponential random variables, which, together with the token probabilities, determine the scores assigned to the tokens. The token with the smallest score wins the race and follows the desired distribution.

Algorithm 1: Token-level sampling (Sun et al., 2023; Chen et al., 2023; Leviathan et al., 2023)

Input: Target and draft distribution P, Q; $X \sim Q$.
Compute the residual P_{res}: for any $x \in \mathcal{V}$,

$$P_{\mathrm{res}}(x) = \frac{P(x) - \min\{P(x), Q(x)\}}{1 - \sum_y \min\{P(y), Q(y)\}}.$$

Sample $U \sim \mathrm{Unif}(0, 1)$
if $U \leq \min(1, P(X)/Q(X))$ **:**
  return $Y = X$ `// Accept`
return $Y \sim P_{\mathrm{res}}$ `// Correction`

Algorithm 2: Token-level sampling by exponential race (Kobus and Gündüz, 2025)

Input: Target and draft distribution P, Q.

for $i \in \mathcal{V}$ **do**
  $E_i \sim \mathrm{Exp}(1)$
$X \leftarrow \arg\min_{i \in \mathcal{V}} E_i / Q(i)$
$Y \leftarrow \arg\min_{i \in \mathcal{V}} E_i / P(i)$
if $X = Y$ **:**
  return $Y = X$ `// Accept`
return Y `// Correction`

4 WATERMARKABLE SPECULATIVE SAMPLING

We first present a single-draft version for illustration. Our use of Poisson processes naturally embeds the watermark into the sampling process via pseudorandomness. The formal multi-draft scheme, which further accelerates autoregressive generation without compromising watermark strength, will be presented in Section 5. Before proceeding, we first introduce two key components of our scheme.

4.1 UNBIASED WATERMARK

A good watermark should be detectable from the generated text while preserving output quality. *Unbiased* schemes use pseudorandomness to couple tokens with secret keys, so that the watermark can be detected through statistical tests without significantly altering the marginal distribution.

Suppose $\tilde{P}$ is a base distribution over $\mathcal{V}$, an unbiased watermark is a decoding rule such that, for each seed ζ, it induces a watermarked distribution $\tilde{P}_\zeta$, which preserves $\tilde{P}$ after averaging over the seed,

$$\mathbf{E}_\zeta[\tilde{P}_\zeta(v)] = \tilde{P}(v), \qquad \forall v \in \mathcal{V}.$$

The Gumbel watermark (Aaronson and Kirchner, 2023), which uses the Gumbel-max trick (Gumbel, 1954) and lets the seed ζ determine the Gumbel noise, is unbiased, and we leverage a similar idea for our scheme. Let $u_t(\cdot|x, y_{t-1}) : \mathcal{V} \to \mathbb{R}$ be a real-valued function (a.k.a. *logits*) that encodes the model's preferences on words and T be the temperature, the softmax sampling is equivalent to

$$y_t = \text{argmax}_{y\in\mathcal{V}}\big(u_t(y)/\mathsf{T}\big) + G_t(y), \quad G_t(y) \sim \text{Gumbel}(0,1) \text{ i.i.d for each } t, y,$$

where the Gumbel noise can be generated by $\text{Gumbel}(0,1) \sim -\log(\log(1/\text{Unif}([0,1])))$.

A random $r_t \sim (\text{Unif}([0,1]))^{|\mathcal{V}|}$ can be represented by $G_t(y) = -\log(-log(r_t(y)))$. We can replace $\text{Unif}([0,1])$ by a pseudo-random function $r_t(y) = F_{y_{t-m:t-1},\mathsf{k}}(y)$ with prefix length m, and employ

$$y_t = \text{argmax}_{y\in\mathcal{V}}\big(u_t(y)/\mathsf{T}\big) - \log(-\log(r_t(y))). \tag{1}$$

Conditioned on a secret key k, $r_t(y)$ is a deterministic function, but over the distribution of k, $r_t(y)$ is computationally indistinguishable from samples drawn from a truly i.i.d. uniform distribution.

The detector has the key k and computes the watermark score $\sum_{t=m+1}^{n} -\log\big(1 - r_t(y_t)\big)$. If $y_{1:n}$ is unwatermarked, the expected score is $n - m$; otherwise, it is larger and the watermark is detected.

Our algorithm relies on Poisson processes and also incorporates the use of pseudo-random functions.

4.2 Poisson Functional Representation (PFR)

We leverage a scheme called PFR (Li and El Gamal, 2018); related works are discussed in Section 2.

Definition 4.1 (PFR)**.** Let $(T_i)_i \sim \text{PP}(1)$ be a Poisson process of rate 1, independent of $Z_i \overset{\text{iid}}{\sim} \nu$ for $i = 1, 2, \ldots$. $(Z_i, T_i)_i$ is a Poisson process of intensity measure $Q \times \lambda_{[0,\infty)}$, where $\lambda_{[0,\infty)}$ is the Lebesgue measure over $[0, \infty)$. Fix distribution P over $\mathcal{Z}$ such that $P \ll Q$. PFR selects the point

$$Z = Z_K \quad \text{where} \quad K := \text{argmin}_i\big(T_i \cdot \big(\mathrm{d}P/\mathrm{d}Q(Z_i)\big)^{-1}\big). \tag{2}$$

Remark 4.2. *By drawing a sequence $(Z_i)_i$ from the* reference *distribution Q and a sequence of* times *$(T_i)_i$, the PFR selects a sample following the* target *distribution P. The sample Z_i with the smallest T_i only follows Q; to obtain a sample from P exactly, we inflate the time by factor $(\mathrm{d}P/\mathrm{d}Q(Z_i))^{-1}$.*

4.3 Watermarkable Speculative Sampling via Poisson Processes

Core idea. We unify watermarking and speculative sampling through a keyed Poisson coupling. Every emitted token is target-distributed, while the draft model only controls how many such samples are emitted. The watermarking is performed at the *coupling level*, rather than through a post-hoc reweighting of marginals. This allows our scheme to stay efficient when embedding watermark. [2]

Algorithm 3 is a sketch of our single-draft speculative sampling with watermark, see a detailed implementation in Algorithm 9. Fixing a Poisson process Π_k by key k; with context c, we denote the labeled Poisson process and the sample selected by PFR by $G(\mathsf{k}, c)$ and $\text{PFR}(\bar{P}(\cdot|c), \Pi)$, respectively.

Note that for discrete LLM target distributions, as we prove in Appendix A.1, PFR reduces to the exponential race (Kobus and Gündüz, 2025). However, Poisson processes become useful, and even necessary, when we extend to the multi-draft version, where we generate multiple samples that all follow the target distribution, by the mapping theorem (Last and Penrose, 2017) for Poisson processes.

He et al. (2026) used a different watermark metric to improve the trade-off frontier. In comparison, our scheme is robust across different watermark measurements, as we consider various scores (Aaronson and Kirchner, 2023; Hu and Huang, 2024; Li et al., 2025; Lattimore, 2026) in Appendix E, which makes our scheme directly comparable to existing watermarking schemes. We use Aaronson's score (Aaronson and Kirchner, 2023) in the main section, which is introduced as follows.

[2] The Maintain Watermark Strength scheme (Hu and Huang, 2024) shares the same spirit; see Figure 1. Nevertheless, our scheme can be extended to the multi-draft version in Section 5, achieving further acceleration.

Algorithm 3: Watermarkable Speculative Sampling via Poisson Processes (Simplified)

Given : lookahead K, output length N, initial prompt $w_{1:n}$, Poisson generator G, secret key k
while $n < N$ **do**
  $h \leftarrow w_{1:n}$; $L \leftarrow \min\{K,\ N-n\}$; *accepted* $\leftarrow$ **true**;
  for $s = 1$ **to** L **do**
    $c_s \leftarrow h \,\|\, \tilde{w}_{1:s-1}$; $\Pi_s \leftarrow G(\mathsf{k}, c_s)$ // keyed Poisson for watermark
    $\tilde{w}_s \leftarrow \mathrm{PFR}(Q(\cdot|c_s), \Pi_s)$
  for $s = 1$ **to** L **do** // evaluation in parallel
    $y_s \leftarrow \mathrm{PFR}(P(\cdot|c_s), \Pi_s)$; $a_s \leftarrow \mathbb{1}\{y_s = \tilde{w}_s\}$; $w_{n+1} \leftarrow y_s$; $n \leftarrow n+1$;
    if $a_s = 0$ **:**
      accepted $\leftarrow$ **false** and **break** // later draft contexts are invalid
  if *accepted* ***and*** $n < N$ **:** // all draft tokens accepted
    $\Pi^\star \leftarrow G(\mathsf{k}, w_{1:n})$ // keyed Poisson source
    $w_{n+1} \leftarrow \mathrm{PFR}(P(\cdot|w_{1:n}), \Pi^\star)$; $n \leftarrow n+1$ // bonus

If the output of key k is $w_{1:N}$ and n_0 is the initial prompt length, the detector evaluates the scored positions $\mathcal{T} := \{n_0+1, \ldots, N\}$. For $t \in \mathcal{T}$, let $c_t := w_{:t-1}$ be the prefix before position t, and the detector regenerates $\Pi_t := G(\mathsf{k}, c_t)$. For each $v \in \mathcal{V}$, let $r_t(v) := \exp(-\mu(v)\tau_t(v))$ be the induced uniform value where $\tau_t(v) \sim \mathrm{Exp}(\mu(v))$ is the first arrival time of v in Π_t, and the detector computes

$$S_{\mathrm{A}} := \sum\nolimits_{t\in\mathcal{T}} -\log(1 - r_t(w_t)), \tag{3}$$

whose expectation is $\mathbf{E}[S_{\mathrm{A}}] \geq |\mathcal{T}| + (\pi^2/6 - 1)\sum_{t\in\mathcal{T}} \mathbf{E}[H(P(\cdot|w_{:t-1}))]$. If the sequence is not generated with k, the observed w_t is independent of Π_t, $r_t(w_t) \sim \mathrm{Unif}[0,1]$ and $\mathbf{E}[S_{\mathrm{A}}] = |\mathcal{T}|$.

At-a-Glance Experimental Validation. We use Figure 1 to illustrate how our algorithm improves the trade-off by increasing sampling efficiency without degrading watermark strength.

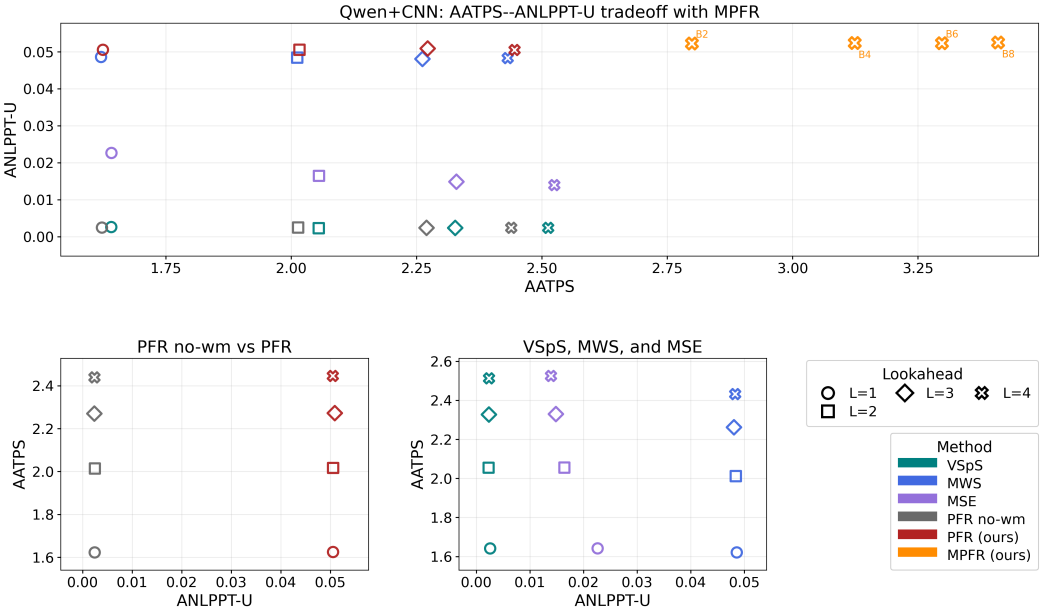


Figure 1: The trade-off between watermark and efficiency on existing methods (Hu and Huang, 2024): Vanilla Speculative Sampling (VSpS), Maintain Watermark Strength (MWS), and Maintain Sampling Efficiency (MSE), and our algorithms (PFR, and multi-draft MPFR that is presented in Section 5). For both PFR (lower-left) and MPFR (upper), we embed watermark (measured by Average Negative Log P-value Per Token (ANLPPT)) without degrading the Average Accepted Tokens Per Step (AATPS). This figure uses Qwen2.5-7B-Instruct and Qwen2.5-0.5B-Instruct as the target and draft, respectively, on dataset CNN_DAILYMAIL. Experiments on more models and datasets are shown in Appendix D.

4.4 DRAFTER INVARIANCE

Conventional speculative sampling suffers from a major disadvantage (Daliri et al., 2025): when the small drafter model $\mathcal{M}_d$ changes, e.g., because of a model update, the tokens generated by the large model may also change due to the coupling (Algorithm 1). In autoregressive language models, this can be even more problematic, since users usually do not want the model output to change under a fixed random seed: *stable* models allow the output to be reliably reproduced for use, testing, and debugging.

Based on the coupling without communication (Bavarian et al., 2016; Li and El Gamal, 2018), *drafter invariant* sampling has recently been studied (Daliri et al., 2025). It refers to the case where, conditioned on the shared randomness $\mathcal{R}$ and the context $x_{:t}$, and potentially other things, the output-prefix law does not depend on the draft model. Let $Y_{1:\tau}, \widetilde{Y}_{1:\tau}$ denote the output sequences from different drafter models in block τ, we define the following notion.

Definition 4.3. The algorithm $\mathcal{A}$ is said to be *stopping-time drafter invariant* if, for every stopping time value τ_0 in the support of τ, every $1 \le j \le \tau_0$, and every output prefix $y_{1:j}$, we have

$$\mathbf{P}\big\{Y_{1:j} = y_{1:j} \,\big|\, \mathcal{R},\, x_{:t},\, \tau = \tau_0\big\} = \mathbf{P}\big\{\widetilde{Y}_{1:j} = y_{1:j} \,\big|\, \mathcal{R},\, x_{:t},\, \widetilde{\tau} = \tau_0\big\}.$$

Proposition 4.4. *Our Algorithm 3 is stopping-time drafter invariant.*

The proof can be found in Appendix C.1.1. The stopping-time drafter invariance is not satisfied by most existing schemes, e.g., SpecTr (Sun et al., 2023), SpecInfer (Miao et al., 2024), and also (Rowan et al., 2025, Algorithm 2).

Intuitively, a stronger notion of drafter invariance helps watermarking but degrades sampling efficiency. The original definition, referred to as the "strong drafter invariance" (Daliri et al., 2025), was shown to have degraded sampling efficiency (Rowan et al., 2025), in where the authors hence proposed a "conditional drafter invariance" that also conditions on the realized draft sequence. We propose the stopping-time drafter invariance that is weaker than the former to admit a better sampling efficiency, but stronger than the latter and more suitable for watermarking. The conditioning on τ is intentional: it separates the role of the drafter in determining the *length* of the current speculative block from its role in determining the *content* of the emitted tokens. Thus, the drafter is allowed to affect the stopping time, and hence the sampling efficiency, but once this stopping time is fixed, the law of every emitted prefix is governed by the target-side randomness and does not depend on the full realized draft sequence.

In watermarking, such invariance is important. A watermark is usually tied to pseudorandomness and is detected from the final output sequence, while speculative sampling introduces an additional ambiguity: an emitted token may come from the draft, the target correction, or the bonus, which affects the watermark statistic, and is one reason behind the no-go trade-off in (Hu and Huang, 2024) and the use of acceptance-side information in (He et al., 2026). However, a watermark detector typically observes only the final tokens and the secret key, not the realized draft sequence or acceptance-side information. Stopping-time drafter invariance is hence better aligned with watermarking: changing the drafter may affect the stopping time and the efficiency, but conditional on the block length, it does not introduce draft-specific dependence into the target-keyed token prefix. This makes the watermark directly tied to the target-side keyed randomness and leads to a more reproducible provenance signal.

Moreover, (He et al., 2026) proposed a new watermark metric that improves the trade-off between watermark strength and sampling efficiency (Hu and Huang, 2024); see, e.g., (He et al., 2026, Figure 1). However, their analysis relies on the maximal coupling probability $\mathbf{P}(X = Y) = 1 - \|P - Q\|_{\mathrm{TV}}$ for coupling $X \sim P, Y \sim Q$ (Thorisson, 2000; Den Hollander, 2012). In contrast, the best matching probability under no-communication coupling is fundamentally different:

$$\mathbf{P}(X = Y) \le (1 - \|P - Q\|_{\mathrm{TV}})/(1 + \|P - Q\|_{\mathrm{TV}}). \tag{4}$$

Thus, enforcing drafter invariance may appear to fundamentally reduce sampling efficiency. We argue, however, that this loss is not inevitable: by using Poisson processes, we derive a multi-draft, drafter-invariant extension that greatly improves the token acceptance rate, as elaborated as follows.

5 WATERMARKABLE MULTI-DRAFT SPECULATIVE SAMPLING

We now extend our Algorithm 3 to multi-draft speculative sampling, by a multi-sample generalization of PFR built on the mapping theorem for Poisson processes (Li and Anantharam, 2021).

We use $\mathcal{M}_t$ and $\mathcal{M}_d^{(b)}$ to denote the *target* and *draft* language models with $1 \le b \le B$. They induce conditional distributions $\mathcal{M}_t(\cdot|x_{:t})$ and $\mathcal{M}_d^{(b)}(\cdot|x_{:t})$, which give the probability of the next token given the current context $x_{:t}$, usually denoted by c. In multi-draft speculative sampling (Miao et al., 2024; Sun et al., 2023; Khisti et al., 2025; Rowan et al., 2025), at each context c, we generate a list of B proposal drafts, often i.i.d. in practice (Rowan et al., 2025; Sun et al., 2023) from $\mathcal{M}_d^{(1)}, \dots, \mathcal{M}_d^{(B)}$. These proposal drafts induce a depth-L speculative tree rooted at the current prefix, which is then verified in parallel with $\mathcal{M}_t$. At each step, if at least one of the B candidate tokens is accepted, the first accepted token will be appended to the final output; otherwise, the verification procedure stops. The output is $Y_{1:\tau}$, where τ is the number of accepted tokens plus one.

The PFR (Definition 4.1) selects a sample $\tilde{Z}_P$ from $(Z_i)_i$ that has the smallest *score* $T_i \cdot \left(\frac{\mathrm{d}P}{\mathrm{d}Q}(Z_i)\right)^{-1}$, where $(T_i)_i \sim \mathrm{PP}(1)$, and it follows the target distribution P. To get B samples, multi-sample PFR (MPFR) selects samples with the smallest, second smallest, and so on up to the B-th smallest scores:

$$\tilde{Z}_P(1), \tilde{Z}_P(2), \dots, \tilde{Z}_P(B) \overset{\text{iid}}{\sim} P, \tag{5}$$

where the exactness follows (Li and Anantharam, 2021); see Appendix A.3 for the formal definition and implementation.

For multi-draft speculative sampling, we use B i.i.d. drafts from a model $\mathcal{M}_d$, and at each sampling step we consider $P := \mathcal{M}_t(\cdot|x_{:t})$ as the target distribution and $Q := \mathcal{M}_d(\cdot|x_{:t})$ as the proposal distribution. Note that we consider *identical* proposals (Sun et al., 2023) in this section, but our framework can also be generalized to non-identically distributed proposals (Rowan et al., 2025), which is deferred to Appendix B.4.

Our scheme is sketched in Algorithm 4; see Algorithm 10 for details. The core idea is as follows. To implement a multi-draft algorithm, Algorithm 4 uses *context-indexed* mapped Poisson processes. Starting from the root prefix $h = w_{1:n}$, we build a depth-ℓ speculative tree, where each occupied context c is assigned a keyed source $\Pi(c)$ and generates $\nu(c)$ drafts from $Q(\cdot|c)$ via MPFR. We then compute the target-side sample $\mathrm{MPFR}(P, \Pi(c), 1)$ for every c and emit tokens by following the unique realized target path, continuing as long as the current target winner belongs to the draft set.

Importantly, the Poisson process is indexed by contexts rather than by drafter identities, and thus MPFR works in a way that makes the emitted tokens tightly tied to the prefix and the secret key, which helps watermark detection. In comparison, (Rowan et al., 2025) uses draft-indexed algorithms, keeps a set of currently viable draft indices, and chooses the next emitted token using only the active drafts. This is a natural implementation, but the active set makes the next emitted token depend on the realized draft sequences, causes the output to retain hidden dependence on draft-side trajectories, and hence only satisfies their conditional drafter invariance, but not stopping-time drafter invariance (Definition 4.3).

Theorem 5.1. *The acceptance probability of at least one token at the current step satisfies*

$$\mathbf{P}(\text{accept}) \ge 1 - \sum_{i \in \mathcal{V}} P(i)\big(P(i)/(P(i) + Q(i))\big)^B. \tag{6}$$

The proof is in Appendix B, where we also give a generalized version that accounts for the number of active drafts, and demonstrate our advantages over the existing schemes (Rowan et al., 2025).

6 EXPERIMENTS

We evaluate whether our algorithms improve the empirical frontier between sampling efficiency and watermark detectability, while preserving generation quality. The headline configuration uses Qwen2.5-7B-Instruct as the target and Qwen2.5-0.5B-Instruct as the drafter on CNN/DAILYMAIL. Other model-dataset cells, full per-L and per-B tables, and ablation details are deferred to Appendix D.

Baselines. We compare against four families of methods, all run in a common harness with the same prompts, decoding parameters, and watermark key. **(i) Watermark-only reference.** BASIC-UWM is autoregressive Gumbel-style unbiased watermarking without speculation (Aaronson and Kirchner, 2023); it sets the watermark-strength ceiling. **(ii) Speculation-only reference.** VSPS

Algorithm 4: Watermarkable Multi-Draft Speculative Sampling (Simplified)

Given : lookahead L, number of homogeneous drafts B, output length N, target model P, draft model Q, initial prompt $w_{1:n}$, keyed source generator G, and secret key k

while $n < N$ **do**

  $h \leftarrow w_{1:n}$; $\ell \leftarrow \min\{L,\, N-n\}$ `// root context of current block`

  initialize root context h with multiplicity B

  for $s = 1$ **to** ℓ **do** `// build context tree up to depth` ℓ

    for *each currently occupied context* c **do**

      $\Pi(c) \leftarrow G(\mathsf{k}, c)$ `// keyed Poisson source for watermark`

      generate $\nu(c)$ drafts by $Q(\cdot|c)$ using $\mathrm{MPFR}(Q(\cdot|c), \Pi(c), \cdot)$ `// Eq.(5)`

      merge equal draft tokens into child contexts and update their multiplicities

  compute target-side sample $y(c) = \mathrm{MPFR}(P(\cdot|c), \Pi(c), 1)$ for every c `// in parallel`

  $c^\star \leftarrow h$; *accepted* $\leftarrow$ **true**

  for $s = 1$ **to** ℓ **do**

    emit the target token $y(c^\star)$ and append it to the output

    if $y(c^\star) \notin \mathcal{D}(c^\star)$ **:** `//` $\mathcal{D}(c^\star)$ `contains distinct draft tokens at` c

      accepted $\leftarrow$ **false** and **break** `// first rejection ends the block`

    if $n = N$ **:**

      break

    $c^\star \leftarrow c^\star \,\|\, y(c^\star)$ `// move to the next realized context`

  if *accepted* ***and*** $n < N$ **:**

    emit one bonus token by $\mathrm{MPFR}(P(\cdot|c^\star), G(\mathsf{k}, c^\star), 1)$

Table 1: Multi-draft results on Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct on CNN/DAILYMAIL and ELI5, $L = 4$. We report the average accepted tokens per step (AATPS) and token rate (TR). The better result between MPFR and GLS for each setting with the same B is in bold. Each cell is computed over the 1000 prompts and 3 different seeds. See more detailed data in Appendix D.4.

		Qwen-CNN/DM		Qwen-ELI5		Llama-CNN/DM		Llama-ELI5	
Decoder	B	AATPS	TR	AATPS	TR	AATPS	TR	AATPS	TR
MPFR	2	2.797	**25.11**	**2.563**	**23.57**	**3.459**	**37.44**	**3.103**	**34.29**
MPFR	4	**3.123**	**27.50**	**2.918**	**26.46**	**3.784**	**40.20**	**3.455**	**37.79**
MPFR	6	**3.294**	**28.50**	**3.109**	**27.81**	**3.935**	**41.71**	**3.635**	**39.59**
MPFR	8	**3.409**	**29.13**	**3.235**	**28.73**	**4.034**	**42.04**	**3.749**	**39.76**
GLS	2	**2.803**	24.78	2.523	22.73	3.431	36.79	3.080	33.72
GLS	4	3.109	27.12	2.850	25.60	3.741	39.31	3.408	37.11
GLS	6	3.272	27.84	3.039	27.01	3.894	40.51	3.588	39.12
GLS	8	3.384	28.62	3.156	28.09	3.987	39.87	3.700	39.37

denotes vanilla speculative sampling without watermark (Leviathan et al., 2023), and the multi-draft list-coupling scheme INVARIANT of Rowan et al. (2025); these set the efficiency ceiling. **(iii) Trade-off frontier.** MWS /MSE of Hu and Huang (2024) are the two endpoints of the no-go trade-off, and MSE-PSEUDO is the pseudorandom-r variant of He et al. (2026) that improves the trade-off via an alternative watermark metric. **(iv) Ours.** PFR (single-draft) and MPFR (multi-draft); PFR-NOWM is PFR with the keyed source replaced by an unkeyed Poisson source, isolating the cost of watermarking from the cost of Poisson coupling.

Metrics. Following (Hu and Huang, 2024; He et al., 2026) we separate three concerns. **Efficiency:** AATPS; see other measures in appendix. **Detectability:** ANLPPT, the average negative log p-value per token under three score variants U / Li / PL (Aaronson and Kirchner, 2023; Li et al., 2025; Lattimore, 2026) (see Appendix E); and TPR@1%FPR, calibrated on non-watermarked generations. **Quality:** per-token log-perplexity (LPPL) under the target model, used as an unbiasedness audit; we also report ROUGE-L when references are available.

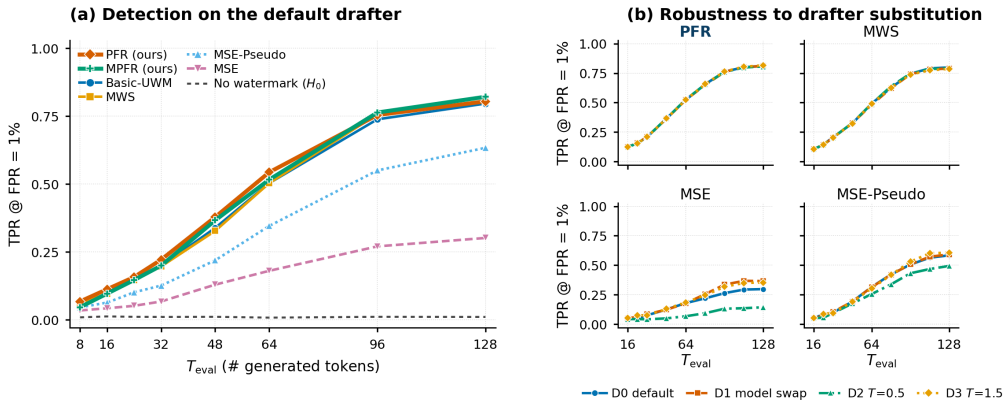


Figure 2: **Detection and robustness to drafter substitution. (a)** TPR@1%FPR versus detection token budget T_{eval} at the default drafter. PFR and MPFR match the strong-watermark references BASIC-UWM and MWS, and outperform MSE/MSE-PSEUDO. **(b)** TPR@1%FPR under drafter substitution: the target, watermark key, and prompts are fixed and only the drafter is swapped. PFR drifts by <1 point, whereas MSE drifts by 12–22 points and MSE-PSEUDO by 6–11 points.

Single-draft efficiency-watermark frontier. Figure 1 plotted every method in the (AATPS, ANLPPT-U) plane sweeping $L \in \{1, 2, 3, 4\}$. The prior speculative watermarking baselines (MWS, MSE, MSE-PSEUDO) exhibit the expected trade-off. PFR dominates both Pareto endpoints at every L. Crucially, PFR and PFR-NOWM have nearly identical AATPS, confirming that the watermark is embedded at the coupling level. Other model/dataset results are tested in Appendix D.3.

Multi-draft scaling. We compare MPFR with the INVARIANT decoder by Rowan et al. (2025) at lookahead L=4 and draft counts $B \in \{2, 4, 6, 8\}$ on the headline cell. Across all values of B, MPFR achieves AATPS comparable to INVARIANT, with only small cell-dependent deviations, while delivering substantially larger ANLPPT-U. The watermark signal is itself stable across B (ΔANLPPT ≤ 0.001), and the per-token detection signal does not dilute as additional drafts are added. Full per-cell numbers across the four (TARGET, DATASET) cells, all three ANLPPT variants, and the LPPL audit column are in Appendix D.4 (Tables 10-13).

Detection at a fixed false-positive rate. ANLPPT measures average evidence per token; TPR@1%FPR measures operational detectability under a calibrated false-positive budget. Figure 2(a) shows TPR@1%FPR versus token budget T_{eval} at the default drafter. PFR and MPFR track the strong-watermark references BASIC-UWM and MWS, while MSE and MSE-PSEUDO are uniformly weaker. MSE-PSEUDO improves over MSE but still falls below our methods, confirming that target-side keyed coupling preserves detector-visible evidence without degrading efficiency.

Drafter-substitution ablation. We test whether the watermark signal is stable to the drafter changes. We vary only the drafter along scale and temperature: D_0 (Qwen2.5-0.5B-Instruct, T=1.0, default), D_1 (Qwen2.5-1.5B-Instruct, T=1.0), D_2 (Qwen2.5-0.5B-Instruct, T=0.5), D_3 (Qwen2.5-0.5B-Instruct, T=1.5). Figure 2(b) shows that our method (PFR) has at most 1 percentage point of TPR drift across D_0–D_3 at $T_{\text{eval}} \in \{64, 128\}$, whereas MSE drifts by 12–22 points and MSE-PSEUDO by 6–11 points. The generation quality (LPPL and ROUGE-L) is verified in Appendix D.5.

CONCLUDING REMARKS AND LIMITATIONS

We designed a novel drafter-invariant, multi-draft speculative sampling algorithm that is watermarkable, improves the trade-off between watermarking and sampling efficiency (Hu and Huang, 2024), and achieves a strong acceleration. We use experiments to verify its strong performance in both aspects. However, a fundamental characterization of the trade-off among target-draft communication, watermark strength, and speculative sampling efficiency remains unclear. Moreover, the current implementation involves substantial coupling and watermark bookkeeping, which slightly slows down the algorithm and may be avoided in the future to further accelerate our algorithms.

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A More on PFR and Related Works

A.1 PFR for Discrete Distributions

It is known that the general PFR (Li and El Gamal, 2018; Li and Anantharam, 2021) reduces to the use of exponential random variables (Li and El Gamal, 2018; Liu and Li, 2025) in the case of simulating discrete distributions, which is similar to the Gumbel-max trick (Gumbel, 1954) used for speculative sampling (Kobus and Gündüz, 2025).

In (Li et al., 2024), it is discussed how to extend the discrete case to the general case. However, the other direction, namely reducing the general case to the discrete case, has not been discussed in the literature, to the best of the authors' knowledge. To make the paper self-contained, we provide formal statements and proofs for this direction.

The first define the exponential functional representation (EFR) as follows (also see (Li and El Gamal, 2018; Liu and Li, 2025)). Let $\mathcal{Z}$ be a finite or countably infinite set, and let P be a distribution on $\mathcal{Z}$. Let $(E_z)_{z\in\mathcal{Z}}$ be independent $\mathrm{Exp}(1)$ random variables. With the convention that $E_z/P(z) = +\infty$ whenever $P(z) = 0$, define

$$Z_{\mathrm{EFR}} := \arg\min_{z\in\mathcal{Z}} \frac{E_z}{P(z)}. \tag{7}$$

As proved in Appendix A.2, EFR is equivalent to the Gumbel-max trick. In the following, we prove that PFR reduces to EFR in the discrete case, i.e., when the target and proposal distributions are discrete.

Theorem A.1. *Let $\mathcal{Z}$ be a finite or countably infinite set. Let Q and P be distributions on $\mathcal{Z}$ such that $P \ll Q$. Write $q_z := Q(z), p_z := P(z), z \in \mathcal{Z}$. The PFR considers a Poisson process $(Z_i, T_i)_{i\geq 1}$ on $\mathcal{Z}\times[0,\infty)$ with intensity measure $Q\times\lambda_{[0,\infty)}$, and $\tilde{T}_i = T_i\left(\frac{\mathrm{d}P}{\mathrm{d}Q}(Z_i)\right)^{-1}$. For each z with $q_z > 0$, define the first arrival time of symbol z by*

$$S_z := \inf\{T_i : Z_i = z\}, \tag{8}$$

and define

$$E_z := q_z S_z. \tag{9}$$

For z with $q_z = 0$, define E_z to be an arbitrary independent $\mathrm{Exp}(1)$ random variable, independent of everything else. Then the random variables $(E_z)_{z\in\mathcal{Z}}$ are independent $\mathrm{Exp}(1)$ random variables, and the PFR output satisfies

$$Z_{\mathrm{PFR}} = \arg\min_i \tilde{T}_i = \arg\min_{z\in\mathcal{Z}} \frac{E_z}{p_z} = Z_{\mathrm{EFR}} \qquad \textit{a.s.}, \tag{10}$$

where we use the convention that $E_z/p_z = +\infty$ if $p_z = 0$.

Proof. For $z \in \mathcal{Z}$ and $t \geq 0$, define

$$N_z(t) := \#\{i : T_i \leq t,\ Z_i = z\}.$$

Since $(Z_i, T_i)_{i\geq 1}$ is a Poisson process with intensity measure $Q\times\lambda_{[0,\infty)}$, for each z with $q_z > 0$, the process $(N_z(t))_{t\geq 0}$ is a Poisson process with rate q_z. Moreover, for distinct symbols $z_1, \ldots, z_m$, the processes

$$(N_{z_1}(t))_{t\geq 0}, \ldots, (N_{z_m}(t))_{t\geq 0}$$

are independent, because they count disjoint subsets of the underlying Poisson process.

Therefore, for z with $q_z > 0$, the first arrival time

$$S_z = \inf\{t : N_z(t) \geq 1\}$$

satisfies

$$\mathbf{P}(S_z > s) = \mathbf{P}(N_z(s) = 0) = e^{-q_z s}, \qquad s \geq 0.$$

Thus $S_z \sim \mathrm{Exp}(q_z)$. Hence

$$E_z = q_z S_z \sim \mathrm{Exp}(1).$$

The independence of the Poisson counting processes for different symbols implies that the random variables

$$(E_z)_{z:q_z>0}$$

are independent. For symbols with $q_z = 0$, we have $p_z = 0$ because $P \ll Q$, and these symbols can never be selected by PFR. Defining independent $\mathrm{Exp}(1)$ variables E_z for those symbols therefore does not change the output. Hence the full family $(E_z)_{z\in\mathcal{Z}}$ is independent and identically distributed as $\mathrm{Exp}(1)$.

It remains to show that the PFR selection rule reduces to the EFR selection rule. Since $P \ll Q$, for every z with $p_z > 0$, we also have $q_z > 0$, and

$$\frac{\mathrm{d}P}{\mathrm{d}Q}(z) = \frac{p_z}{q_z}.$$

Therefore, for every index i with $Z_i = z$ and $p_z > 0$,

$$\tilde{T}_i = T_i \left(\frac{p_z}{q_z}\right)^{-1} = T_i \frac{q_z}{p_z}.$$

Taking the minimum over all Poisson points with mark z, we get

$$\inf_{i:Z_i=z} \tilde{T}_i = \left(\inf_{i:Z_i=z} T_i\right) \frac{q_z}{p_z} = S_z \frac{q_z}{p_z} = \frac{E_z}{p_z}.$$

If $p_z = 0$, then $\mathrm{d}P/\mathrm{d}Q(z) = 0$ whenever $q_z > 0$, and hence

$$\tilde{T}_i = T_i \left(\frac{\mathrm{d}P}{\mathrm{d}Q}(z)\right)^{-1} = +\infty$$

for every Poisson point with mark z. This agrees with the EFR convention

$$\frac{E_z}{p_z} = +\infty.$$

If $q_z = 0$, then also $p_z = 0$, and no Poisson point has mark z almost surely, so such a symbol is irrelevant in both constructions.

Thus,

$$\inf_i \tilde{T}_i = \inf_{z\in\mathcal{Z}} \inf_{i:Z_i=z} \tilde{T}_i = \inf_{z\in\mathcal{Z}} \frac{E_z}{p_z}.$$

Moreover, the minimizer is unique almost surely. Indeed, for every z with $p_z > 0$,

$$\frac{E_z}{p_z} \sim \mathrm{Exp}(p_z),$$

and these variables are independent. Since exponential random variables are continuous, ties have probability zero. Therefore, almost surely,

$$Z_{\mathrm{PFR}} = Z_K = \arg\min_{z\in\mathcal{Z}} \frac{E_z}{p_z} = Z_{\mathrm{EFR}}.$$

□

A.2 Relation between PFR and Gumbel-Max Trick

As discussed in (Zhao et al., 2025a), the Gumbel Watermark (Aaronson and Kirchner, 2023) is based on

$$y_t = \mathrm{argmin}_{y\in\mathcal{V}} \frac{u_t(y)}{T} + G_t(y)$$

where $G(y) \sim \text{Gumbel}(0,1)$ i.i.d. for each t and y. Here, we show that it is equivalent to the PFR in the discrete case. Recall $\text{Gumbel}(0,1) \sim -\log\big(\log(1/\text{Unif}[0,1])\big)$. We calculate

$$\begin{aligned}
y_t &= \arg\max_{y\in\mathcal{V}} \left(\frac{u_t(y)}{T} + G_t(y)\right) \\
&= \arg\max_{y\in\mathcal{V}} \left(\frac{u_t(y)}{T} - \log Z_t(y)\right) \qquad \Big(Z_t(y) \overset{\text{iid}}{\sim} \text{Exp}(1),\ G_t(y) \overset{d}{=} -\log Z_t(y)\Big) \\
&= \arg\max_{y\in\mathcal{V}} \Big(-\log\big(Z_t(y)e^{-u_t(y)/T}\big)\Big) \\
&= \arg\min_{y\in\mathcal{V}} Z_t(y)e^{-u_t(y)/T} \\
&= \arg\min_{y\in\mathcal{V}} \frac{Z_t(y)}{\exp(u_t(y)/T)} \\
&= \arg\min_{y\in\mathcal{V}} \frac{Z_t(y)}{w_t(y)} \qquad \Big(w_t(y) := \exp(u_t(y)/T)\Big) \\
&= \arg\min_{y\in\mathcal{V}} \frac{Z_t(y)}{P_t(y)},
\end{aligned}$$

where the last equality is because $P_t(y) := \frac{w_t(y)}{\sum_{y'\in\mathcal{V}} w_t(y')}$ and scaling by a positive constant doesn't change the argmin.

A.3 Mapped Poisson Process

A.3.1 Definitions

Definition A.2 (Mapped Poisson Process (Li and Anantharam, 2021))**.** Let $(T_i)_i$ be a Poisson process of rate 1, independent of $Z_i \overset{\text{iid}}{\sim} Q$ for $i = 1, 2, \ldots$. Let $i_{P,1}, i_{P,2}, \ldots \in \mathbb{N}$ be a sequence of distinct integers such that $\bigcup_{j=1}^{\infty} \{i_{P,j}\} = \{i : \frac{\mathrm{d}P}{\mathrm{d}Q}(Z_i) > 0\}$ and $\big\{T_{i_{P,j}}\big(\frac{\mathrm{d}P}{\mathrm{d}Q}(Z_{i_{P,j}})\big)^{-1}\big\}_{j\in\mathbb{N}}$ is sorted in ascending order with arbitrary tie-breaking.[3] For $j \in \mathbb{N}$ and suppose P is the target distribution, we denote

$$\left\{\tilde{Z}_P(j), \tilde{T}(j)\right\}_{j\in\mathbb{N}} := \left\{Z_{i_{P,j}}, T_{i_{P,j}}\big(\frac{\mathrm{d}P}{\mathrm{d}Q}(Z_{i_{P,j}})\big)^{-1}\right\}_{j\in\mathbb{N}}. \tag{11}$$

Give a positive integer B, the mapped Poisson Process selects the first B samples in $\left\{\tilde{Z}_P(j), \tilde{T}(j)\right\}_{j\in\mathbb{N}}$.

Proposition A.3. *By the mapping theorem (Last and Penrose, 2017), $\left\{\tilde{Z}_P(j), \tilde{T}(j)\right\}_{j\in\mathbb{N}}$ is a Poisson process with intensity measure $P \times \lambda_{\mathbb{R}_{\geq 0}}$ where $\mathbb{R}_{\geq 0}$ to denote nonnegative real numbers, and therefore we have*

$$\tilde{Z}_P(1), \tilde{Z}_P(2), \ldots \overset{\text{iid}}{\sim} P. \tag{12}$$

A.3.2 Direct Implementations

We can implement the MPFR just like (Theis and Ahmed, 2022) for PFR, as follows:

[3] Note a tie occurs with probability 0.

Algorithm 5: Finite implementation of the B-sample mapped PFR

Given : target distribution P, proposal distribution Q, likelihood ratio $r = dP/dQ$, number of samples B, and a constant $\underline{w} > 0$ such that $1/r(u) \geq \underline{w}$ Q-a.s. on $\{r(u) > 0\}$

Return : B samples from P

$H \leftarrow \emptyset$ `// H stores the current B smallest-score candidates`

$t \leftarrow 0$; $i \leftarrow 1$

while $|H| < B$ ***or*** $\max_{(s,j,z)\in H} s > t\underline{w}$ **do**

  Sample $E_i \sim \mathrm{Exp}(1)$ and set $t \leftarrow t + E_i$ `// Poisson process`

  Sample $Z_i \sim Q$

  if $r(Z_i) > 0$ **:**

    $S_i \leftarrow t/r(Z_i)$

  else

    $S_i \leftarrow +\infty$

  if $|H| < B$ **:**

    Insert (S_i, i, Z_i) into H

  else if $S_i < \max_{(s,j,z)\in H} s$ **:**

    Remove an element $(s^\star, j^\star, z^\star) \in H$ satisfying $s^\star = \max_{(s,j,z)\in H} s$

    Insert (S_i, i, Z_i) into H

  $i \leftarrow i + 1$

Sort the elements of H as

$$(S_{i_1}, i_1, Z_{i_1}), \ldots, (S_{i_B}, i_B, Z_{i_B})$$

so that

$$S_{i_1} \leq S_{i_2} \leq \cdots \leq S_{i_B}.$$

return $(Z_{i_1}, \ldots, Z_{i_B})$

A.3.3 Proof of Correctness

We then prove that the implementation of Algorithm 5 is correct.

Theorem A.4. *Let $P \ll Q$ be probability measures on a Polish space $\mathcal{U}$, and let $r := dP/dQ$. Let $Z_1, Z_2, \ldots \overset{iid}{\sim} Q$ and let $0 < T_1 < T_2 < \cdots$ be the arrival times of a unit-rate Poisson process, independent of $(Z_i)_{i\geq 1}$. Define*

$$S_i := \begin{cases} T_i/r(Z_i), & r(Z_i) > 0, \\ +\infty, & r(Z_i) = 0. \end{cases}$$

Assume that there exists $\underline{w} > 0$ such that

$$1/r(u) \geq \underline{w} \qquad Q\text{-a.s. on } \{r(u) > 0\}.$$

Fix $B \in \mathbb{N}$. Consider the algorithm that scans the candidates in the order $i = 1, 2, \ldots$, keeps the B smallest observed scores, and stops at the first time n such that the current B-th smallest observed score is at most $T_n\underline{w}$. Then the algorithm terminates almost surely after finitely many iterations. Moreover, at termination, the retained candidates are exactly the B smallest-score candidates among the infinite sequence. Consequently, the returned marks are distributed as

$$(\widetilde{U}_P(1), \ldots, \widetilde{U}_P(B)),$$

the first B points of the mapped Poisson process with respect to P, and hence are i.i.d. according to P.

Proof. Let

$$\Pi_Q := \{(Z_i, T_i) : i \geq 1\}.$$

Since $0 < T_1 < T_2 < \cdots$ are the arrival times of a unit-rate Poisson process and $Z_1, Z_2, \ldots \overset{iid}{\sim} Q$ are independent marks, Π_Q is a Poisson point process on $\mathcal{U} \times \mathbb{R}_{\geq 0}$ with intensity measure $Q \times \lambda$, where λ denotes the Lebesgue measure on $\mathbb{R}_{\geq 0}$.

Consider the measurable mapping

$$\Phi : \mathcal{U} \times \mathbb{R}_{\geq 0} \to \mathcal{U} \times \mathbb{R}_{\geq 0}, \qquad \Phi(u, t) = \left(u, \frac{t}{r(u)}\right)$$

on the set $\{u : r(u) > 0\}$. Points with $r(u) = 0$ are assigned score $+\infty$ and hence never appear among the finite mapped arrival times. For any nonnegative measurable function g on $\mathcal{U} \times \mathbb{R}_{\geq 0}$, we have

$$\begin{aligned}\int_{\mathcal{U}} \int_0^\infty g\left(u, \frac{t}{r(u)}\right) dt\, Q(du) &= \int_{\{r>0\}} \int_0^\infty g\left(u, \frac{t}{r(u)}\right) dt\, Q(du) \\ &= \int_{\{r>0\}} \int_0^\infty g(u,s) r(u)\, ds\, Q(du) \\ &= \int_{\mathcal{U}} \int_0^\infty g(u,s)\, ds\, P(du).\end{aligned}$$

Therefore, by the mapping theorem for Poisson point processes, the mapped point process

$$\Pi_P := \{(Z_i, S_i) : i \geq 1,\ r(Z_i) > 0\}, \qquad S_i := \frac{T_i}{r(Z_i)},$$

is a Poisson point process on $\mathcal{U} \times \mathbb{R}_{\geq 0}$ with intensity measure $P \times \lambda$.

Let

$$S_{i_1} < S_{i_2} < \cdots$$

be the increasing ordering of the finite scores in Π_P. Since the time-marginal intensity of Π_P is λ, the ordered times $S_{i_1}, S_{i_2}, \ldots$ are the arrival times of a unit-rate Poisson process. Moreover, conditional on these arrival times, the associated marks are independent with common distribution P. Hence

$$(Z_{i_1}, Z_{i_2}, \ldots) \stackrel{d}{=} (U_1, U_2, \ldots),$$

where $U_1, U_2, \ldots \overset{iid}{\sim} P$. In particular,

$$(Z_{i_1}, \ldots, Z_{i_B})$$

are exactly the first B points of the mapped Poisson process with respect to P, and they are i.i.d. according to P.

It remains to show that the finite algorithm stops safely and terminates almost surely. After observing candidates $1, \ldots, n$, let

$$S_{n,(B)}$$

denote the B-th smallest value among $S_1, \ldots, S_n$, whenever at least B finite scores have been observed. By construction, the algorithm keeps the B smallest observed scores, so the largest score retained by the algorithm is exactly $S_{n,(B)}$.

Suppose the algorithm stops at time n. Then

$$S_{n,(B)} \leq T_n \underline{w}.$$

For every future candidate $m > n$, we have $T_m > T_n$. If $r(Z_m) = 0$, then $S_m = +\infty$, so such a candidate cannot improve the current list. If $r(Z_m) > 0$, then by the assumption $1/r(u) \geq \underline{w}$ Q-a.s.,

$$S_m = \frac{T_m}{r(Z_m)} = T_m \frac{1}{r(Z_m)} \geq T_m \underline{w} > T_n \underline{w} \geq S_{n,(B)}.$$

Thus every future score is strictly larger than the current B-th smallest observed score. Therefore no future candidate can enter the set of the global B smallest scores. Hence, when the algorithm stops, the retained candidates are exactly the global B smallest-score candidates among the infinite sequence.

Finally, we prove almost sure finite termination. Since Π_P is a Poisson point process with intensity $P \times \lambda$, its first B mapped arrival times are finite almost surely. Let

$$S_{i_B}$$

denote the B-th smallest mapped score, and let

$$N_0 := \max\{i_1, \ldots, i_B\}.$$

Then $N_0 < \infty$ almost surely. After the algorithm has processed candidates $1, \ldots, N_0$, it has already observed the true global first B mapped candidates, so the largest retained score is S_{i_B}. Since

$$T_n \to \infty \qquad \text{almost surely},$$

and since $\underline{w} > 0$, there exists an almost surely finite random index $N \geq N_0$ such that

$$T_N \underline{w} \geq S_{i_B}.$$

At time N, the algorithm therefore satisfies the stopping condition

$$S_{N,(B)} = S_{i_B} \leq T_N \underline{w}.$$

Thus the algorithm terminates almost surely after finitely many iterations.

Combining the safe stopping argument with the mapped Poisson process argument, the algorithm returns exactly

$$(Z_{i_1}, \ldots, Z_{i_B}),$$

which are the first B points of the mapped Poisson process with respect to P. Consequently, the returned samples are i.i.d. according to P. □

A.4 A BETTER FINITE-SUPPORT MPFR IMPLEMENTATION

The proposal-scanning implementation in Appendix A.3.2 can be slow in finite-vocabulary LLM decoding. Its stopping rule depends on a lower bound on the likelihood ratio, which can be very small when a rare token has small proposal probability but non-negligible target probability. In our experiments, the logits are processed by temperature and top-k truncation, so the resulting distribution has finite support of size at most K. In this setting, we can simulate the relevant Poisson clocks directly and avoid proposal scanning altogether.

Let P be a distribution on a finite vocabulary $\mathcal{U}$, and define

$$\mathcal{A}_P := \{u \in \mathcal{U} : P(u) > 0\}.$$

For each $u \in \mathcal{A}_P$, let

$$G_{u,j} := \sum_{m=1}^{j} E_{u,m}, \qquad j \geq 1,$$

where the $E_{u,m}$'s are independent $\mathrm{Exp}(1)$ random variables. We associate to (u, j) the mapped-Poisson score

$$S_{u,j} := \frac{G_{u,j}}{P(u)}.$$

The first B MPFR samples are the labels of the B smallest scores among

$$\{S_{u,j} : u \in \mathcal{A}_P,\ j \geq 1\}.$$

Since no token can contribute more than B arrivals among the first B total arrivals, it is enough to generate $j = 1, \ldots, B$ for every $u \in \mathcal{A}_P$.

Theorem A.5 (Exactness of direct finite-support MPFR)**.** *Let P be a distribution on a finite set $\mathcal{U}$. The output $(U_{(1)}, \ldots, U_{(B)})$ of Algorithm 6 consists of B independent samples from P. That is, for every $(u_1, \ldots, u_B) \in \mathcal{U}^B$,*

$$\mathbf{P}\{U_{(1)} = u_1, \ldots, U_{(B)} = u_B\} = \prod_{b=1}^{B} P(u_b).$$

Proof. For each $u \in \mathcal{A}_P$, the sequence $(G_{u,j})_{j \geq 1}$ is the arrival process of a unit-rate Poisson process. Therefore,

$$\left(\frac{G_{u,j}}{P(u)}\right)_{j \geq 1}$$

is a Poisson process on $\mathbb{R}_+$ with rate $P(u)$, since

$$\#\left\{j : \frac{G_{u,j}}{P(u)} \leq t\right\} = \#\{j : G_{u,j} \leq P(u)t\}$$

is Poisson with mean $P(u)t$. The processes corresponding to different tokens are independent. Their superposition is therefore a marked Poisson process on $\mathbb{R}_+ \times \mathcal{U}$ with intensity measure

$$dt \otimes P.$$

Algorithm 6: Direct finite-support implementation of B-sample MPFR

Given : finite distribution P on $\mathcal{U}$, number of samples B, and keyed source Π
Return : B samples from P
$\mathcal{A}_P \leftarrow \{u \in \mathcal{U} : P(u) > 0\}$
$H \leftarrow \emptyset$ `// candidate triples` (S, u, j)
for *each* $u \in \mathcal{A}_P$ **do**
  $G \leftarrow 0$
  for $j = 1$ **to** B **do**
    Generate $E_{u,j} \sim \mathrm{Exp}(1)$ from Π
    $G \leftarrow G + E_{u,j}$
    $S \leftarrow G/P(u)$
    Insert (S, u, j) into H
Let

$$(S_{(1)}, U_{(1)}, J_{(1)}), \ldots, (S_{(B)}, U_{(B)}, J_{(B)})$$

be the B elements of H with smallest scores, ordered by increasing score
return $(U_{(1)}, \ldots, U_{(B)})$

Equivalently, the total arrival rate is $\sum_{u \in \mathcal{A}_P} P(u) = 1$, and the marks of the ordered arrivals are independent with common distribution P.

Algorithm 6 constructs the first B candidate arrivals of each token clock. This is sufficient because among the first B arrivals of the superposed process, no individual token clock can contribute more than B arrivals. Hence the B smallest elements in H are exactly the first B arrivals of the superposed marked Poisson process, and their labels are B independent samples from P. □

We now describe the batched speculative decoding implementation. For a context c, let $P_k(\cdot|c)$ and $Q_k(\cdot|c)$ denote the processed target and draft distributions after applying the same logits processing, such as temperature and top-k truncation. The draft side builds a tree of candidate continuations using Algorithm 6 with Q_k. The target side then evaluates all contexts needed for verification in a batched manner and uses the same MPFR clocks with P_k to follow the realized target path. This batching changes only the order and grouping of model evaluations, not the sampling law.

Theorem A.6 (Exactness of batched-target direct-top-k MPFR decoding)**.** *Assume that for every context c, the keyed source $\Pi_{\mathsf{k}}(c)$ provides independent exponential clocks across tokens and local arrival indices, and that the sources $\{\Pi_{\mathsf{k}}(c)\}_c$ are independent across distinct contexts. Then Algorithm 7 generates exactly from the autoregressive model with transition kernel $P_k(\cdot|c)$. That is, for every finite sequence $u_1, \ldots, u_m$,*

$$\mathbf{P}\{W_{n+1} = u_1, \ldots, W_{n+m} = u_m\} = \prod_{t=1}^{m} P_k(u_t | w_{1:n}, u_1, \ldots, u_{t-1}).$$

Proof. Fix a context c. By Theorem A.5,

$$Y(c) = \mathrm{MPFR}(P_k(\cdot|c), \Pi_{\mathsf{k}}(c), 1)$$

has distribution $P_k(\cdot|c)$. The draft samples generated from $Q_k(\cdot|c)$ may use the same keyed clocks, and hence are coupled with $Y(c)$, but this does not change the marginal distribution of $Y(c)$.

Let

$$C_t := (w_{1:n}, W_{n+1}, \ldots, W_{n+t-1})$$

be the random context before generating W_{n+t}. The event $\{C_t = c\}$ is determined by the keyed sources associated with strict prefix contexts of c. By the assumed independence across contexts, $\Pi_{\mathsf{k}}(c)$ is independent of this event. Therefore, conditional on $C_t = c$,

$$W_{n+t} = Y(c) \sim P_k(\cdot|c),$$

and hence

$$\mathbf{P}\{W_{n+t} = u | C_t = c\} = P_k(u|c).$$

Algorithm 7: Batched-target direct-top-k MPFR speculative sampling

Given : lookahead L, number of drafts B, output length N, target model P, draft model Q, prompt $w_{1:n}$, key k

while $n < N$ **do**

$h \leftarrow w_{1:n}$, $\ell \leftarrow \min\{L, N-n\}$

$\mathcal{C}_0 \leftarrow \{h\}$ and $\nu(h) \leftarrow B$ `// context multiplicities`

`// Draft tree construction`

for $s = 1$ **to** ℓ **do**

$\mathcal{C}_s \leftarrow \emptyset$

for *each* $c \in \mathcal{C}_{s-1}$ **do**

Generate

$$(X_1(c), \ldots, X_{\nu(c)}(c)) \leftarrow \text{MPFR}(Q_k(\cdot|c), \Pi_{\mathsf{k}}(c), \nu(c)).$$

Let $\mathcal{D}(c)$ be the set of distinct tokens among these samples

for *each* $u \in \mathcal{D}(c)$ **do**

$m(c,u) \leftarrow |\{i : X_i(c) = u\}|$

Add $c\|u$ to $\mathcal{C}_s$

$\nu(c\|u) \leftarrow \nu(c\|u) + m(c,u)$

`// Batched target evaluation`

Compute the processed target distributions $P_k(\cdot|c)$ for all contexts

$$c \in \bigcup_{s=0}^{\ell} \mathcal{C}_s$$

using batched target-model forward passes

for *each such context* c **do**

Generate

$$Y(c) \leftarrow \text{MPFR}(P_k(\cdot|c), \Pi_{\mathsf{k}}(c), 1).$$

`// Follow the realized target path`

$c^\star \leftarrow h$, accepted $\leftarrow$ **true**

for $s = 1$ **to** ℓ **do**

Emit $Y(c^\star)$ and append it to the output

if $Y(c^\star) \notin \mathcal{D}(c^\star)$ **:**

accepted $\leftarrow$ **false**

break `// first rejection ends the block`

if $n = N$ **:**

break

$c^\star \leftarrow c^\star\|Y(c^\star)$

if accepted ***and*** $n < N$ **:**

Emit the precomputed bonus token $Y(c^\star)$

Applying the chain rule gives

$$\begin{aligned}
&\mathbf{P}\{W_{n+1} = u_1, \ldots, W_{n+m} = u_m\} \\
&\quad = \prod_{t=1}^{m} \mathbf{P}\{W_{n+t} = u_t | W_{n+1} = u_1, \ldots, W_{n+t-1} = u_{t-1}\} \\
&\quad = \prod_{t=1}^{m} P_k(u_t | w_{1:n}, u_1, \ldots, u_{t-1}).
\end{aligned}$$

Thus the generated sequence has exactly the desired autoregressive law. Finally, batched target evaluation only changes the computational order of deterministic model evaluations. It does not change the context-indexed MPFR clocks or the resulting samples $Y(c)$, so it does not change the output distribution. □

This implementation removes the proposal-scanning overhead by simulating the token-wise Poisson clocks directly. For support size at most K, producing B MPFR samples requires KB exponential variables followed by a top-B selection. The batched target implementation further reduces the number of target-model calls: instead of verifying the realized path sequentially with roughly one target forward pass per generated token, it evaluates the contexts in a speculative block together and extracts all logits needed for verification. If a block emits BE tokens on average, the target-forward cost per emitted token is reduced from approximately 1 to approximately $1/\mathrm{BE}$, while preserving the exact target marginal distribution.

B Theoretic Guarantees

B.1 Proof of Theorem 5.1

Proof. The proof is based on the Poisson matching lemma (Li and Anantharam, 2021). Let $p_i := P(i)$ and $q_i := Q(i)$ for $i \in \mathcal{V}$. Let $Y := \tilde{Z}_P(1)$ and $X^{(b)} := \tilde{Z}_Q(b)$ for $b = 1, \ldots, B$ be generated from the same mapped Poisson process, and define the token-level acceptance event

$$\mathcal{A}_t := \left\{ Y \in \{X^{(1)}, \ldots, X^{(B)}\} \right\},$$

which is the acceptance event. Apply the (Li and Anantharam, 2021, Lemma 3) with $j = 1$ and $k = B$. Conditioned on $Y = \tilde{Z}_P(1) = i$, the event $\mathcal{A}_t^c$ is exactly the event that the first mapped P-sample does not appear among the first B mapped Q-samples. Hence

$$\mathbf{P}(\mathcal{A}_t^c | Y = i) \le \left(1 - \left(1 + \frac{p_i}{q_i}\right)^{-1}\right)^B = \left(\frac{p_i}{p_i + q_i}\right)^B.$$

Taking complements gives

$$\mathbf{P}(\mathcal{A}_t | Y = i) \ge 1 - \left(\frac{p_i}{p_i + q_i}\right)^B.$$

Finally, average over $Y \sim P$:

$$\mathbf{P}(\mathcal{A}_t) = \sum_{i \in \mathcal{V}} p_i \mathbf{P}(\mathcal{A}_t | Y = i) \ge \sum_{i \in \mathcal{V}} p_i \left[1 - \left(\frac{p_i}{p_i + q_i}\right)^B\right].$$

For $B = 1$, this simplifies to

$$1 - \sum_{i \in \mathcal{V}} \frac{p_i^2}{p_i + q_i} = \sum_{i \in \mathcal{V}} \frac{p_i q_i}{p_i + q_i}.$$ □

B.2 Comparison with (Rowan et al., 2025) on Acceptance Probability

We compare our token-level acceptance probability with the token-level acceptance probability from (Rowan et al., 2025) in two levels, and we prove that in each level our algorithm's acceptance probability is strictly better.

B.2.1

We first prove a stronger version of Theorem 5.1, but not as clean as it.

Theorem B.1. *Let P be the target distribution and Q be the draft distribution on a finite vocabulary $\mathcal{V}$. Assume first that $P(i), Q(i) > 0$ for all $i \in \mathcal{V}$. For each $i \in \mathcal{V}$, define*

$$\alpha_i := P(i) \sum_{j \in \mathcal{V}} \left(\frac{Q(j)}{Q(i)} - \frac{P(j)}{P(i)} \right)_+ .$$

If the target token is generated as $\tilde{U}_P(1)$ and the B draft tokens are generated as $\tilde{U}_Q(1), \ldots, \tilde{U}_Q(B)$ from the same underlying Poisson process, then the acceptance probability of at least one token satisfies

$$\mathbf{P}(\text{accept}) \geq 1 - \sum_{i \in \mathcal{V}} P(i) \left(\frac{\alpha_i}{1 + \alpha_i} \right)^B . \tag{13}$$

Proof. Let the target token be

$$Y := \tilde{U}_P(1),$$

and let the B draft tokens be

$$X_b := \tilde{U}_Q(b), \qquad b = 1, \ldots, B.$$

The acceptance event is

$$\text{accept} = \{Y \in \{X_1, \ldots, X_B\}\}.$$

Conditioning on $Y = i$, the failure event is

$$\{Y \notin \{X_1, \ldots, X_B\}\}.$$

By the exact $j = 1$ tail formula in the generalized Poisson matching lemma,

$$\mathbf{P}(Y \notin \{X_1, \ldots, X_B\} \,|\, Y = i) \leq \left(\frac{\alpha_i}{1 + \alpha_i} \right)^B ,$$

where

$$\alpha_i = P(i) \sum_{j \in \mathcal{V}} \left(\frac{Q(j)}{Q(i)} - \frac{P(j)}{P(i)} \right)_+ .$$

Therefore,

$$\mathbf{P}(\text{accept} | Y = i) \geq 1 - \left(\frac{\alpha_i}{1 + \alpha_i} \right)^B .$$

Averaging over $Y \sim P$ gives

$$\begin{aligned} \mathbf{P}(\text{accept}) &= \sum_{i \in \mathcal{V}} P(i) \mathbf{P}(\text{accept} | Y = i) \\ &\geq \sum_{i \in \mathcal{V}} P(i) \left[1 - \left(\frac{\alpha_i}{1 + \alpha_i} \right)^B \right] \\ &= 1 - \sum_{i \in \mathcal{V}} P(i) \left(\frac{\alpha_i}{1 + \alpha_i} \right)^B . \end{aligned}$$

This proves (13). □

Moreover,

$$1 - \sum_{i \in \mathcal{V}} P(i) \left(\frac{\alpha_i}{1 + \alpha_i} \right)^B \geq 1 - \sum_{i \in \mathcal{V}} P(i) \left(\frac{P(i)}{P(i) + Q(i)} \right)^B . \tag{14}$$

Thus (13) strictly improves the weaker bound

$$\mathbf{P}(\text{accept}) \geq 1 - \sum_{i \in \mathcal{V}} P(i) \left(\frac{P(i)}{P(i) + Q(i)} \right)^B$$

whenever the inequality in (14) is strict.

We then show above Theorem B.1 is tighter than (Rowan et al., 2025, Proposition 2).

Theorem B.2. *Let P be the target distribution and Q be the draft distribution on a finite vocabulary $\mathcal{V}$. Write*

$$p_i := P(i), \qquad q_i := Q(i), \qquad i \in \mathcal{V},$$

and assume first that $p_i, q_i > 0$ for all $i \in \mathcal{V}$. For each $i \in \mathcal{V}$, define

$$\alpha_i := p_i \sum_{j \in \mathcal{V}} \left(\frac{q_j}{q_i} - \frac{p_j}{p_i} \right)_+ .$$

Let

$$L_{\mathrm{PML}} := 1 - \sum_{i \in \mathcal{V}} p_i \left(\frac{\alpha_i}{1 + \alpha_i} \right)^B$$

be the acceptance lower bound obtained from the exact $j = 1$ tail formula of the generalized Poisson matching lemma. Let

$$L_{\mathrm{LML}} := \sum_{i \in \mathcal{V}} \frac{B}{\sum_{j \in \mathcal{V}} \left[\max\left\{ \frac{p_j}{p_i}, \frac{q_j}{q_i} \right\} + (B-1)\frac{p_j}{p_i} \right]}$$

be the lower bound in Eq. (3) of the list matching lemma, written in the same target–draft notation. Then

$$L_{\mathrm{PML}} \geq L_{\mathrm{LML}}.$$

Moreover, if $B > 1$ and $P \neq Q$, then

$$L_{\mathrm{PML}} > L_{\mathrm{LML}}.$$

Proof. The proof is based on the Poisson matching lemma (Li and Anantharam, 2021). Let $Y := \tilde{Z}_P(1)$ and $X^{(b)} := \tilde{Z}_Q(b)$ for $b = 1, \ldots, B$ be generated from the same mapped Poisson process. By the mapping theorem, $\tilde{Z}_Q(1), \tilde{Z}_Q(2), \ldots$ are i.i.d. according to Q, and $\tilde{Z}_P(1) \sim P$. Define the token-level acceptance event

$$\mathcal{A}_t := \left\{ Y \in \{X^{(1)}, \ldots, X^{(B)}\} \right\}.$$

Apply the exact $j = 1$ tail formula of the generalized Poisson matching lemma with the first distribution equal to P and the second distribution equal to Q. Conditioned on $Y = \tilde{Z}_P(1) = i$, the event $\mathcal{A}_t^c$ is the event that the first mapped P-sample does not appear among the first B mapped Q-samples. Hence

$$\mathbf{P}(\mathcal{A}_t^c | Y = i) \leq \left(\frac{\alpha_i}{1 + \alpha_i} \right)^B,$$

where

$$\alpha_i = p_i \sum_{j \in \mathcal{V}} \left(\frac{q_j}{q_i} - \frac{p_j}{p_i} \right)_+ .$$

Taking complements gives

$$\mathbf{P}(\mathcal{A}_t | Y = i) \geq 1 - \left(\frac{\alpha_i}{1 + \alpha_i} \right)^B.$$

Finally, averaging over $Y \sim P$ gives

$$\begin{aligned} \mathbf{P}(\mathcal{A}_t) &= \sum_{i \in \mathcal{V}} p_i \mathbf{P}(\mathcal{A}_t | Y = i) \\ &\geq \sum_{i \in \mathcal{V}} p_i \left[1 - \left(\frac{\alpha_i}{1 + \alpha_i} \right)^B \right] \\ &= 1 - \sum_{i \in \mathcal{V}} p_i \left(\frac{\alpha_i}{1 + \alpha_i} \right)^B = L_{\mathrm{PML}}. \end{aligned}$$

We now rewrite the LML bound in terms of the same quantities α_i. For each fixed $i \in \mathcal{V}$,

$$\begin{aligned} p_i \sum_{j \in \mathcal{V}} \max\left\{\frac{p_j}{p_i}, \frac{q_j}{q_i}\right\} &= \sum_{j \in \mathcal{V}} \max\left\{p_j, \frac{p_i}{q_i} q_j\right\} \\ &= \sum_{j \in \mathcal{V}} \left[p_j + \left(\frac{p_i}{q_i} q_j - p_j\right)_+\right] \\ &= 1 + p_i \sum_{j \in \mathcal{V}} \left(\frac{q_j}{q_i} - \frac{p_j}{p_i}\right)_+ \\ &= 1 + \alpha_i. \end{aligned}$$

Therefore,

$$\begin{aligned} \sum_{j \in \mathcal{V}} \left[\max\left\{\frac{p_j}{p_i}, \frac{q_j}{q_i}\right\} + (B-1)\frac{p_j}{p_i}\right] &= \frac{1+\alpha_i}{p_i} + (B-1)\frac{1}{p_i} \\ &= \frac{B + \alpha_i}{p_i}. \end{aligned}$$

Thus Eq. (3) of the list matching lemma can be written as

$$L_{\mathrm{LML}} = \sum_{i \in \mathcal{V}} p_i \frac{B}{B + \alpha_i}.$$

It remains to compare the two expressions term by term. For every $\alpha \geq 0$, we claim that

$$1 - \left(\frac{\alpha}{1+\alpha}\right)^B \geq \frac{B}{B+\alpha}.$$

If $\alpha = 0$, both sides are equal to 1. If $\alpha > 0$, the inequality is equivalent to

$$\left(\frac{\alpha}{1+\alpha}\right)^B \leq \frac{\alpha}{B+\alpha},$$

or equivalently

$$\left(1 + \frac{1}{\alpha}\right)^B \geq 1 + \frac{B}{\alpha}.$$

This follows immediately from the binomial theorem:

$$\left(1 + \frac{1}{\alpha}\right)^B = 1 + \frac{B}{\alpha} + \sum_{m=2}^{B} \binom{B}{m} \alpha^{-m} \geq 1 + \frac{B}{\alpha}.$$

Hence, for each $i \in \mathcal{V}$,

$$p_i \left[1 - \left(\frac{\alpha_i}{1+\alpha_i}\right)^B\right] \geq p_i \frac{B}{B+\alpha_i}.$$

Summing over $i \in \mathcal{V}$ yields

$$L_{\mathrm{PML}} \geq L_{\mathrm{LML}}.$$

Finally, suppose $B > 1$ and $P \neq Q$. We show that the inequality is strict. If $\alpha_i = 0$ for every $i \in \mathcal{V}$, then for every $i, j \in \mathcal{V}$,

$$\frac{q_j}{q_i} \leq \frac{p_j}{p_i}.$$

Equivalently,

$$\frac{q_j}{p_j} \leq \frac{q_i}{p_i}.$$

Since this holds for every pair (i, j), all ratios q_i/p_i must be equal to a common constant. Because both P and Q are probability distributions, this constant must be 1, and hence $P = Q$, a contradiction. Therefore, when $P \neq Q$, there exists at least one i such that $\alpha_i > 0$.

For $B > 1$ and $\alpha > 0$, the binomial expansion above is strict:

$$\left(1 + \frac{1}{\alpha}\right)^B > 1 + \frac{B}{\alpha}.$$

Hence

$$1 - \left(\frac{\alpha}{1+\alpha}\right)^B > \frac{B}{B+\alpha}.$$

Applying this to an index i with $\alpha_i > 0$ gives a strict term in the sum. Therefore

$$L_{\mathrm{PML}} > L_{\mathrm{LML}}.$$

This completes the proof. □

B.2.2

We then show the clean version, Theorem 5.1, is already tighter than (Rowan et al., 2025, Eq. (4)).

Theorem B.3. *Let P be the target distribution and Q be the draft distribution on a finite vocabulary $\mathcal{V}$. Write*

$$p_i := P(i), \qquad q_i := Q(i), \qquad i \in \mathcal{V}.$$

Assume first that $p_i, q_i > 0$ for all $i \in \mathcal{V}$. Define

$$L_{\mathrm{PML,weak}} := 1 - \sum_{i \in \mathcal{V}} p_i \left(\frac{p_i}{p_i + q_i}\right)^B$$

and

$$L_{\mathrm{LML,rel}} := \sum_{i \in \mathcal{V}} p_i \left(1 + \frac{p_i}{Bq_i}\right)^{-1} = \sum_{i \in \mathcal{V}} p_i \frac{Bq_i}{p_i + Bq_i}.$$

Then

$$L_{\mathrm{PML,weak}} \geq L_{\mathrm{LML,rel}}.$$

Moreover, if $B > 1$, then the inequality is strict.

Proof. The proof is based on the Poisson matching lemma (Li and Anantharam, 2021). Let $Y := \tilde{Z}_P(1)$ and let

$$X^{(b)} := \tilde{Z}_Q(b), \qquad b = 1, \ldots, B,$$

be generated from the same mapped Poisson process. Define the token-level acceptance event

$$\mathcal{A}_t := \left\{ Y \in \{X^{(1)}, \ldots, X^{(B)}\} \right\}.$$

Applying the relaxed Poisson matching bound with $j = 1$ and $k = B$, conditioned on $Y = \tilde{Z}_P(1) = i$, gives

$$\mathbf{P}(\mathcal{A}_t^c | Y = i) \leq \left(1 - \left(1 + \frac{p_i}{q_i}\right)^{-1}\right)^B = \left(\frac{p_i}{p_i + q_i}\right)^B.$$

Taking complements yields

$$\mathbf{P}(\mathcal{A}_t | Y = i) \geq 1 - \left(\frac{p_i}{p_i + q_i}\right)^B.$$

Averaging over $Y \sim P$ gives

$$\begin{aligned}
\mathbf{P}(\mathcal{A}_t) &= \sum_{i \in \mathcal{V}} p_i \mathbf{P}(\mathcal{A}_t | Y = i) \\
&\geq \sum_{i \in \mathcal{V}} p_i \left[1 - \left(\frac{p_i}{p_i + q_i}\right)^B\right] \\
&= 1 - \sum_{i \in \mathcal{V}} p_i \left(\frac{p_i}{p_i + q_i}\right)^B = L_{\mathrm{PML,weak}}.
\end{aligned}$$

We now compare this lower bound with the relaxed version of the list matching lemma. In the notation of the list matching lemma paper, the proposal distribution is denoted by p and the target distribution by q. Since here we use P for the target and Q for the draft, their relaxed conditional bound, Eq. (4), becomes

$$\mathbf{P}(\mathcal{A}_t|Y=i) \geq \left(1+\frac{p_i}{Bq_i}\right)^{-1} = \frac{Bq_i}{p_i+Bq_i}.$$

Thus the corresponding unconditional relaxed LML bound is

$$L_{\mathrm{LML,rel}} = \sum_{i\in\mathcal{V}} p_i \frac{Bq_i}{p_i+Bq_i}.$$

It remains to compare the two conditional terms. Let

$$r_i := \frac{p_i}{q_i}.$$

Then

$$1-\left(\frac{p_i}{p_i+q_i}\right)^B = 1-\left(\frac{r_i}{1+r_i}\right)^B,$$

whereas

$$\frac{Bq_i}{p_i+Bq_i} = \frac{B}{B+r_i}.$$

Hence it suffices to prove that, for every $r>0$,

$$1-\left(\frac{r}{1+r}\right)^B \geq \frac{B}{B+r}.$$

Equivalently,

$$\left(\frac{r}{1+r}\right)^B \leq \frac{r}{B+r}.$$

Since $r>0$, this is equivalent to

$$\left(1+\frac{1}{r}\right)^B \geq 1+\frac{B}{r}.$$

This follows from the binomial theorem:

$$\left(1+\frac{1}{r}\right)^B = 1+\frac{B}{r}+\sum_{m=2}^{B}\binom{B}{m}r^{-m} \geq 1+\frac{B}{r}.$$

Therefore, for each $i\in\mathcal{V}$,

$$1-\left(\frac{p_i}{p_i+q_i}\right)^B \geq \frac{Bq_i}{p_i+Bq_i}.$$

Multiplying by p_i and summing over $i\in\mathcal{V}$ gives

$$L_{\mathrm{PML,weak}} \geq L_{\mathrm{LML,rel}}.$$

Finally, if $B>1$, then for every $r>0$,

$$\sum_{m=2}^{B}\binom{B}{m}r^{-m} > 0.$$

Thus the above inequality is strict for every i with $p_i, q_i > 0$. Under full support, all such terms have positive weight p_i, and hence

$$L_{\mathrm{PML,weak}} > L_{\mathrm{LML,rel}}.$$

This completes the proof. □

B.3 Generalized Theoretic Guarantee and Proof

Theorem B.4. *Consider the exact draft-indexed active-set version of multi-draft speculative sampling with total draft count B. Fix a current decoding step t, and let $S_t \subseteq \{1, \dots, B\}$ be the active set before sampling the next token. Write $J_t := |S_t|$. Let the current realized context be c_t, and similarly define $P_t := \mathcal{M}_t(\cdot|c_t), Q_t := \mathcal{M}_d(\cdot|c_t)$. For $i \in \mathcal{V}$, write $p_i^{(t)} := P_t(i)$ and $q_i^{(t)} := Q_t(i)$. Then the token-level acceptance probability at step t satisfies*

$$\mathbf{P}(\mathcal{A}_t|c_t, S_t) \geq \sum_{i\in\mathcal{V}} p_i^{(t)} \left[1 - \left(1 - \left(1 + \frac{p_i^{(t)}}{q_i^{(t)}}\right)^{-1}\right)^{J_t}\right] = 1 - \sum_{i\in\mathcal{V}} p_i^{(t)} \left(\frac{p_i^{(t)}}{p_i^{(t)} + q_i^{(t)}}\right)^{J_t}.$$

In particular, at the first step of each speculative block one has $J_t = B$, so

$$\mathbf{P}(\mathcal{A}_t|c_t) \geq 1 - \sum_{i\in\mathcal{V}} p_i^{(t)} \left(\frac{p_i^{(t)}}{p_i^{(t)} + q_i^{(t)}}\right)^{B}.$$

Proof. Conditioned on the realized context c_t and the active set S_t, the exact active-set algorithm uses only the J_t active draft streams for both target-side selection and acceptance. Inactive drafts do not participate in the current-step minimum and therefore do not affect the current-step coupling.

Hence, after conditioning on (c_t, S_t), the current step is identical to the setting of Theorem 5.1 with the number of drafts equal to J_t. Applying Theorem 5.1 with B replaced by J_t gives

$$\mathbf{P}(\mathcal{A}_t|c_t, S_t) \geq 1 - \sum_{i\in\mathcal{V}} p_i^{(t)} \left(\frac{p_i^{(t)}}{p_i^{(t)} + q_i^{(t)}}\right)^{J_t}.$$

At the first step of a speculative block, all drafts are active, so $J_t = B$, which yields the final claim. □

B.4 Extension of Theorem B.1 to non-identically distributed proposals

Proposition B.5. *Let P be the target distribution on a finite vocabulary $\mathcal{V}$, and let $Q_1, \dots, Q_B$ be B proposal distributions. Write*

$$p_i := P(i), \qquad q_{b,i} := Q_b(i), \qquad i \in \mathcal{V},\ b \in [B].$$

Assume first that $p_i, q_{b,i} > 0$ for all i and b. For each $i \in \mathcal{V}$ and $b \in [B]$, define

$$\alpha_i^{(b)} := p_i \sum_{j\in\mathcal{V}} \left(\frac{q_{b,j}}{q_{b,i}} - \frac{p_j}{p_i}\right)_+.$$

Then there exists a common-randomness coupling such that $Y \sim P$, $X^{(b)} \sim Q_b$ for each $b \in [B]$, the proposals $X^{(1)}, \dots, X^{(B)}$ are mutually independent, and

$$\Pr\Big[Y \in \{X^{(1)}, \dots, X^{(B)}\} \,\Big|\, Y = i\Big] \geq \sum_{b=1}^{B} \frac{1}{B + \alpha_i^{(b)}}. \tag{15}$$

Consequently,

$$\Pr\Big[Y \in \{X^{(1)}, \dots, X^{(B)}\}\Big] \geq \sum_{i\in\mathcal{V}} p_i \sum_{b=1}^{B} \frac{1}{B + \alpha_i^{(b)}}. \tag{16}$$

Proof. The proof is based on the Poisson matching lemma (Li and Anantharam, 2021). The main difference from the identical-proposal case is that there is no single proposal ordering when the proposal laws are $Q_1, \dots, Q_B$. We therefore augment the sample space by a block label. Define

$$\widetilde{\mathcal{V}} := \mathcal{V} \times [B],$$

and introduce the augmented target and proposal distributions

$$\widetilde{P}(i,b) := \frac{p_i}{B}, \qquad \widetilde{Q}_b(i,\ell) := \mathbf{1}\{\ell = b\} q_{b,i}.$$

Let

$$(Y, J) := \tilde{Z}_{\widetilde{P}}(1), \qquad (X^{(b)}, b) := \tilde{Z}_{\widetilde{Q}_b}(1), \quad b = 1, \dots, B,$$

be generated from the same underlying Poisson process on $\widetilde{\mathcal{V}} \times \mathbb{R}_{\geq 0}$. By the mapping theorem, $(Y, J) \sim \widetilde{P}$, and hence $Y \sim P$ and J is uniform on $[B]$ and independent of Y. Also, for each b, $(X^{(b)}, b) \sim \widetilde{Q}_b$, so $X^{(b)} \sim Q_b$. Since the laws $\widetilde{Q}_b$ are supported on disjoint blocks, the corresponding proposal samples are mutually independent.

Fix $i \in \mathcal{V}$. Define the acceptance event

$$\mathcal{A} := \left\{ Y \in \{X^{(1)}, \dots, X^{(B)}\} \right\}.$$

For each $b \in [B]$, let

$$\mathcal{A}_b := \left\{ (Y, J) = (X^{(b)}, b) \right\}.$$

Then $\mathcal{A} \supseteq \bigcup_{b=1}^{B} \mathcal{A}_b$, and the events $\mathcal{A}_1, \dots, \mathcal{A}_B$ are pairwise disjoint because they occur in different blocks. Therefore,

$$\begin{aligned}
\mathbf{P}(\mathcal{A} | Y = i) &\geq \sum_{b=1}^{B} \mathbf{P}(\mathcal{A}_b | Y = i) \\
&= \sum_{b=1}^{B} \mathbf{P}(J = b | Y = i) \mathbf{P}(\mathcal{A}_b | Y = i, J = b) \\
&= \frac{1}{B} \sum_{b=1}^{B} \Pr\left[\tilde{Z}_{\widetilde{P}}(1) = \tilde{Z}_{\widetilde{Q}_b}(1) \,\middle|\, \tilde{Z}_{\widetilde{P}}(1) = (i,b) \right]. \qquad (17)
\end{aligned}$$

We now apply the exact $j = 1$ tail formula in the generalized Poisson matching lemma to the pair $(\widetilde{P}, \widetilde{Q}_b)$. Conditioned on $\tilde{Z}_{\widetilde{P}}(1) = (i, b)$, the local PML parameter is

$$\begin{aligned}
\widetilde{\alpha}_i^{(b)} &:= \widetilde{P}(i,b) \sum_{\ell=1}^{B} \sum_{j \in \mathcal{V}} \left(\frac{\widetilde{Q}_b(j,\ell)}{\widetilde{Q}_b(i,b)} - \frac{\widetilde{P}(j,\ell)}{\widetilde{P}(i,b)} \right)_+ \\
&= \frac{p_i}{B} \sum_{j \in \mathcal{V}} \left(\frac{q_{b,j}}{q_{b,i}} - \frac{p_j}{p_i} \right)_+ \\
&= \frac{\alpha_i^{(b)}}{B}.
\end{aligned}$$

Hence

$$\Pr\left[\tilde{Z}_{\widetilde{P}}(1) = \tilde{Z}_{\widetilde{Q}_b}(1) \,\middle|\, \tilde{Z}_{\widetilde{P}}(1) = (i,b) \right] \geq 1 - \frac{\widetilde{\alpha}_i^{(b)}}{1 + \widetilde{\alpha}_i^{(b)}} = \frac{B}{B + \alpha_i^{(b)}}.$$

Substituting this into (17) gives

$$\mathbf{P}(\mathcal{A} | Y = i) \geq \frac{1}{B} \sum_{b=1}^{B} \frac{B}{B + \alpha_i^{(b)}} = \sum_{b=1}^{B} \frac{1}{B + \alpha_i^{(b)}}.$$

This proves (15). Averaging over $Y \sim P$ gives (16). □

C Drafter Invariance

C.1 Definitions

For a speculative sampling algorithm $\mathcal{A}$ with target model $\mathcal{M}_t$, draft model $\mathcal{M}_d$, shared randomness $\mathcal{R}$ and context $x_{:t}$, let $Y_{1:\tau}$ denote the emitted output sequence of $\mathcal{A}$ in the current speculative block.

Definition C.1 ((Daliri et al., 2025))**.** The algorithm $\mathcal{A}$ is said to be *strongly drafter invariant* if for every $1 \le j \le \tau$,

$$\mathbf{P}\big\{Y_{1:j} = y_{1:j} \,\big|\, \mathcal{R},\, x_{:t}\big\} = \mathbf{P}\big\{\widetilde{Y}_{1:j} = y_{1:j} \,\Big|\, \mathcal{R},\, x_{:t}\big\}.$$

Definition C.2 ((Rowan et al., 2025))**.** The algorithm $\mathcal{A}$ is said to be *conditional drafter invariant* if for every $1 \le j \le \tau$, letting $X_{1:L} := X_{1:L}(\mathcal{M}_d)$ denote the length-L draft sequence by $\mathcal{M}_d$, we have

$$\mathbf{P}\big\{Y_{1:j} = y_{1:j} \,\big|\, \mathcal{R},\, x_{:t},\, X_{1:L} = x_{1:L}\big\} = \mathbf{P}\big\{\widetilde{Y}_{1:j} = y_{1:j} \,\big|\, \mathcal{R},\, x_{:t},\, \widetilde{X}_{1:L} = x_{1:L}\big\}.$$

C.1.1 Proof of Proposition 4.4

We rephrase Proposition 4.4 to a more detailed version as follows.

Proposition C.3 (Stopping-time drafter invariance)**.** *Let τ denote the number of tokens emitted in one speculative block of Algorithm 3. Fix the initial prompt $h := w_{1:n}$ and the shared randomness*

$$\mathcal{R} := \{\, G(\mathsf{k}, c) : c \text{ is any context} \,\}.$$

Define the target-side PFR chain *recursively by*

$$Z_1 := \mathrm{PFR}\big(P(\cdot|h),\, G(\mathsf{k}, h)\big),$$

and for $s \ge 2$,

$$Z_s := \mathrm{PFR}\big(P(\cdot|h\|Z_{1:s-1}),\, G(\mathsf{k},\, h\|Z_{1:s-1})\big).$$

Then, for any stopping-time value τ_0 and every $1 \le j \le \tau_0$,

$$Y_{1:j} = Z_{1:j} \qquad \text{almost surely on the event } \{\tau = \tau_0\}.$$

Equivalently, for every output prefix $y_{1:j}$,

$$\mathbf{P}\big\{Y_{1:j} = y_{1:j} \,\big|\, \mathcal{R},\, h,\, \tau = \tau_0\big\} = \mathbb{1}\{Z_{1:j} = y_{1:j}\}.$$

In particular, for any two draft models Q and $\widetilde{Q}$,

$$\mathbf{P}\big\{Y_{1:j} = y_{1:j} \,\big|\, \mathcal{R},\, h,\, \tau = \tau_0\big\} = \mathbf{P}\big\{\widetilde{Y}_{1:j} = y_{1:j} \,\big|\, \mathcal{R},\, h,\, \widetilde{\tau} = \tau_0\big\},$$

so Algorithm 3 is stopping-time drafter invariant in the sense of Definition 4.3.

Proof. Fix $(\mathcal{R}, h)$ and a stopping-time value τ_0. We first show by induction on s that, on the event $\{\tau \ge s\}$,

$$Y_s = Z_s.$$

For $s = 1$, the first context is

$$c_1 = h.$$

Hence

$$Y_1 = \mathrm{PFR}\big(P(\cdot|c_1),\, G(\mathsf{k}, c_1)\big) = \mathrm{PFR}\big(P(\cdot|h),\, G(\mathsf{k}, h)\big) = Z_1.$$

Now suppose $s \ge 2$ and that $Y_r = Z_r$ holds on $\{\tau \ge r\}$ for all $r = 1, \dots, s-1$. On the event $\{\tau \ge s\}$, the block has reached step s, so the first $s-1$ draft tokens were accepted. Therefore

$$\tilde{w}_{1:s-1} = Y_{1:s-1}.$$

By the induction hypothesis, on $\{\tau \ge s\}$ we also have

$$Y_{1:s-1} = Z_{1:s-1},$$

and hence

$$c_s = h\|\tilde{w}_{1:s-1} = h\|Y_{1:s-1} = h\|Z_{1:s-1}.$$

Thus,

$$Y_s = \mathrm{PFR}\big(P(\cdot|c_s),\, G(\mathsf{k}, c_s)\big) = \mathrm{PFR}\big(P(\cdot|h\|Z_{1:s-1}),\, G(\mathsf{k},\, h\|Z_{1:s-1})\big) = Z_s.$$

This completes the induction.

Now let $1 \le j \le \tau_0$. On the event $\{\tau = \tau_0\}$ we have $\tau \ge j$, so the above induction gives

$$Y_{1:j} = Z_{1:j} \qquad \text{almost surely on } \{\tau = \tau_0\}.$$

Therefore,

$$\mathbf{P}\big\{Y_{1:j} = y_{1:j} \,\big|\, \mathcal{R},\, h,\, \tau = \tau_0\big\} = \mathbb{1}\{Z_{1:j} = y_{1:j}\}.$$

Since the right-hand side depends only on $(\mathcal{R}, h, \tau_0)$ and not on the draft model, the same conditional law holds for any other draft model $\widetilde{Q}$ conditioned on $\widetilde{\tau} = \tau_0$. This is exactly stopping-time drafter invariance. □

C.2 MULTI-DRAFT DRAFTER INVARIANCE

For multi-draft speculative sampling scheme $\mathcal{A}$ with target model $\mathcal{M}_t$, draft models $\mathcal{M}_d^{(1)}, \ldots, \mathcal{M}_d^{(B)}$, context $x_{:t}$, shared randomness $\mathcal{R}$ and $b \in [B]$, let $X_{1:L}^{(b)} := X_{1:L}(\mathcal{M}_d^{(b)})$ denote the length-L draft sequence by $\mathcal{M}_d^{(b)}$ and $Y_{1:\tau}$ denote the emitted output sequence of $\mathcal{A}$ in the current speculative block.

Definition C.4 (Strong drafter invariance)**.** The algorithm $\mathcal{A}$ is *drafter invariant* if for every $1 \le j \le \tau$, every initial context $x_{:t}$, every realization of the shared randomness $\mathcal{R}$, and every output prefix $y_{1:j}$,

$$\mathbf{P}\Big\{Y_{1:j} = y_{1:j} \,\Big|\, \mathcal{R},\, x_{:t}\Big\} = \mathbf{P}\Big\{\widetilde{Y}_{1:j} = y_{1:j} \,\Big|\, \mathcal{R},\, x_{:t}\Big\},$$

for any two choices of draft models

$$\mathcal{M}_d^{(1)}, \ldots, \mathcal{M}_d^{(B)} \quad \text{and} \quad \widetilde{\mathcal{M}}_d^{(1)}, \ldots, \widetilde{\mathcal{M}}_d^{(B)}.$$

Definition C.5 (Conditional drafter invariance)**.** We say that the algorithm is *conditionally drafter invariant* if, for every $1 \le j \le \tau$, every initial context $x_{:t}$, every realization of the shared randomness $\mathcal{R}$, every draft sequences $x_{1:L}^{(1)}, \ldots, x_{1:L}^{(B)}$, and every output prefix $y_{1:j}$,

$$\begin{aligned}&\mathbf{P}\Big\{Y_{1:j} = y_{1:j} \,\Big|\, \mathcal{R},\, x_{:t},\, X_{1:L}^{(1)} = x_{1:L}^{(1)}, \ldots, X_{1:L}^{(B)} = x_{1:L}^{(B)}\Big\}\\ &= \mathbf{P}\Big\{\widetilde{Y}_{1:j} = y_{1:j} \,\Big|\, \mathcal{R},\, x_{:t},\, \widetilde{X}_{1:L}^{(1)} = x_{1:L}^{(1)}, \ldots, \widetilde{X}_{1:L}^{(B)} = x_{1:L}^{(B)}\Big\},\end{aligned}$$

for any two choices of draft models

$$\mathcal{M}_d^{(1)}, \ldots, \mathcal{M}_d^{(B)} \quad \text{and} \quad \widetilde{\mathcal{M}}_d^{(1)}, \ldots, \widetilde{\mathcal{M}}_d^{(B)}.$$

That is, conditioned on the shared randomness, the initial context, and the realized draft sequences, the conditional law of the emitted output prefix does not depend on which draft models generated those sequences.

Definition C.6 (stopping time drafter invariance)**.** Consider a speculative sampling algorithm with target model $\mathcal{M}_t$, draft models $\mathcal{M}_d^{(1)}, \ldots, \mathcal{M}_d^{(B)}$, initial context $x_{:t}$, shared randomness $\mathcal{R}$, and block output $Y_{1:\tau}$, where τ is the number of emitted tokens in the current speculative block.

We say that the algorithm is *stopping time drafter invariant* if, for every $1 \le j \le \tau$, every initial context $x_{:t}$, every realization of the shared randomness $\mathcal{R}$, every block length value τ_0, and every output prefix $y_{1:j}$,

$$\begin{aligned}&\mathbf{P}\Big\{Y_{1:j} = y_{1:j} \,\Big|\, \mathcal{R},\, x_{:t},\, \tau = \tau_0\Big\}\\ &= \mathbf{P}\Big\{\widetilde{Y}_{1:j} = y_{1:j} \,\Big|\, \mathcal{R},\, x_{:t},\, \widetilde{\tau} = \tau_0\Big\},\end{aligned}$$

for any two choices of draft models

$$\mathcal{M}_d^{(1)}, \ldots, \mathcal{M}_d^{(B)} \quad \text{and} \quad \widetilde{\mathcal{M}}_d^{(1)}, \ldots, \widetilde{\mathcal{M}}_d^{(B)}.$$

That is, conditioned on the shared randomness, the initial context, and the realized block length, the conditional law of the emitted output prefix does not depend on the draft models.

D EXPERIMENT DETAILS AND FULL RESULTS

This appendix expands the main-text evaluation along the two axes that mirror the contributions of Section 1: (i) the *single-draft regime* (B=1), where we certify that our keyed Poisson sampler preserves acceptance and quality while carrying watermark signal; and (ii) the *multi-draft regime* (B>1), where we extend the unbiasedness/strength guarantees to larger B and benchmark sampling efficiency against the list-coupling scheme of Rowan et al. (2025).

D.1 SETUP

Models. We use two target/drafter pairs that span the two model families used in the prior watermark and list-coupling literature. The headline configuration follows Rowan et al. (2025): target `Qwen2.5-7B-Instruct` with drafter `Qwen2.5-0.5B-Instruct`. For instruction-following diagnostics we additionally use `lmsys/vicuna-7b-v1.5` paired with the small same-family drafter `double7/vicuna-68m`; the FastChat v1.1 chat template is injected explicitly because the released checkpoint does not ship a `chat_template` attribute. All models run in `float16` on a single NVIDIA H100 GPU.

Datasets. We evaluate on two tasks chosen to match the coverage of prior work: CNN/DAILYMAIL (summarization, prompted with the `System:/INPUT:/OUTPUT:` template of Hu and Huang (2024)) and ELI5 (open-ended explanation, Fan et al. (2019)). Each cell uses n=1000 prompts.

Sampling parameters. Unless stated otherwise we use top_k=50, top_p=1.0, temperature T=1.0, and `max_new_tokens`=128. The watermark key is fixed across all runs so that detection is deterministic.

Sweeps. For single-draft we sweep lookahead $L \in \{1, 2, 3, 4\}$, matching Hu and Huang (2024). For multi-draft we fix L=4 and sweep draft counts $B \in \{2, 4, 6, 8\}$, matching Rowan et al. (2025). Each $(\text{MODEL}, \text{DATASET}, \text{DECODER}, L, B)$ configuration is run with multiple independent seeds; tables in Appendices D.3–D.4 report $\text{mean} \pm \text{std}$ across the 1000 prompts of the cell.

Baselines. We adopt the four-family taxonomy used in the main text (Section 6). For convenience we restate the role of each baseline:

- **Quality anchor** — BASIC-UWM: autoregressive Gumbel-style unbiased watermarking with no speculation. Provides the LPPL reference and the watermark-strength ceiling.
- **Speculation-only references** — VSPS (Leviathan et al., 2023) at B=1, and the INVARIANT (*InvariantMultiDraftStrategy*) of Rowan et al. (2025) at B>1. These set the efficiency ceiling.
- **Trade-off frontier** — MWS and MSE (Hu and Huang, 2024), plus MSE-PSEUDO, the dual-key pseudorandom-r variant of He et al. (2026). These mark the Pareto frontier we aim to circumvent.
- **Ours** — PFR (single-draft), MPFR (multi-draft), and the unwatermarked variant PFR-NOWM.

Metrics. We restate the three metric families used in the main text. *Efficiency.* **AATPS** (average accepted tokens per step) and **Token rate** in tokens/s. The latter is hardware-dependent and is reported only as a diagnostic. *Detectability.* **ANLPPT** — the average negative log p-value per token — under three score variants: **ANLPPT-U** (Aaronson and Kirchner, 2023; Hu and Huang, 2024), **ANLPPT-Li** (Li et al., 2025), and **ANLPPT-PL** (Lattimore, 2026). We additionally report **TPR@1%FPR** based on the Aaronson Gamma-tail test (Appendix D.5, "Detection statistic"). *Quality.* **LPPL** (per-token log-perplexity under the target), reported as an audit column. We require LPPL parity with BASIC-UWM to certify unbiasedness, and we do not optimise against this metric.

Reproducibility. The exact $(\text{seed}, \text{MODEL}, \text{DATASET}, \text{DECODER}, L, B)$ configurations, software versions (`torch`, `transformers`, attention kernel), GPU model, and per-seed result files are released with the paper.

D.2 PROMPT TEMPLATES

For all instruction-tuned targets we apply the model's native chat template via the HuggingFace `apply_chat_template` interface. The Vicuna-v1.5 release ships without a `chat_template` attribute, so we explicitly inject the FastChat v1.1 template before tokenization.

CNN/DailyMail (summarization).

```
[
  {"role": "system",
   "content": "You are a helpful assistant."},
  {"role": "user",
   "content": "Summarize the following article in 3-5 sentences:\n\n"
              + article[:1500]}
]
```

The article is truncated to 1500 characters before tokenization.

ELI5 (open-ended QA).

```
[
  {"role": "system",
   "content": "You are a helpful assistant."},
  {"role": "user",
   "content": "Please explain like I'm five:\n\n" + question}
]
```

For each prompt the model generates up to 128 new tokens under the sampling parameters of Appendix D.1. The watermark key is fixed across all runs.

D.3 SINGLE-DRAFT EXPERIMENTS: FULL RESULTS

Goal. Hu and Huang (2024) prove that, under speculative sampling, any unbiased reweighting scheme must give up either watermark strength (MSE-style) or acceptance rate (MWS-style); their Theorem 3.1 forbids schemes that strictly improve both axes simultaneously. We claim that our Poisson-process construction sidesteps the trade-off.

Protocol. We follow Hu and Huang (2024): the same metric battery (AATPS, ANLPPT, LPPL) and the same lookahead range $L \in \{1, 2, 3, 4\}$. We evaluate on `Qwen2.5-7B-Instruct/0.5B` and on `vicuna-7b-v1.5/vicuna-68m`, on both CNN/DAILYMAIL and ELI5. On top of the Hu–Huang decoder set we add (i) our PFR sampler, (ii) the unwatermarked variant PFR-NOWM, (iii) BASIC-UWM as the unbiasedness anchor, and (iv) the pseudorandom-r variant MSE-PSEUDO of He et al. (2026).

Headline figure. Figure 1 (in Section 4) plots every method as a curve in the $(\text{AATPS}, \text{ANLPPT-}U)$ plane sweeping L on the headline cell. Figures 3–5 report the same plot on the remaining three cells. PFR dominates the MWS/MSE Pareto frontier on every cell: at every L it sits both higher (greater detection signal) and to the right (greater acceptance) than the trade-off baselines.

Unbiasedness audit. We report LPPL under the target model as an empirical audit of whether the keyed Poisson coupling introduces a measurable likelihood shift. PFR matches BASIC-UWM in LPPL within statistical noise, as shown in the final column of Tables 2–5. Together with the AATPS and ANLPPT results, this supports our empirical claim that coupling-level watermarking preserves both acceptance behavior and detector-visible signal; the exact marginal guarantee follows from the PFR construction.

Per-cell tables. Tables 2–5 report all metrics per $(L, \text{DECODER})$ for the four $(\text{TARGET}, \text{DATASET})$ cells. Each cell reports $\text{mean} \pm \text{std}$ across the 1000 prompts of the cell.

D.4 MULTI-DRAFT EXPERIMENTS: FULL PER-CELL RESULTS

Setup. The main-text multi-draft paragraph (Section 6, "Multi-draft scaling") summarises a single representative cell (Qwen2.5-7B-Instruct $\times$ CNN/DAILYMAIL). Here we extend the

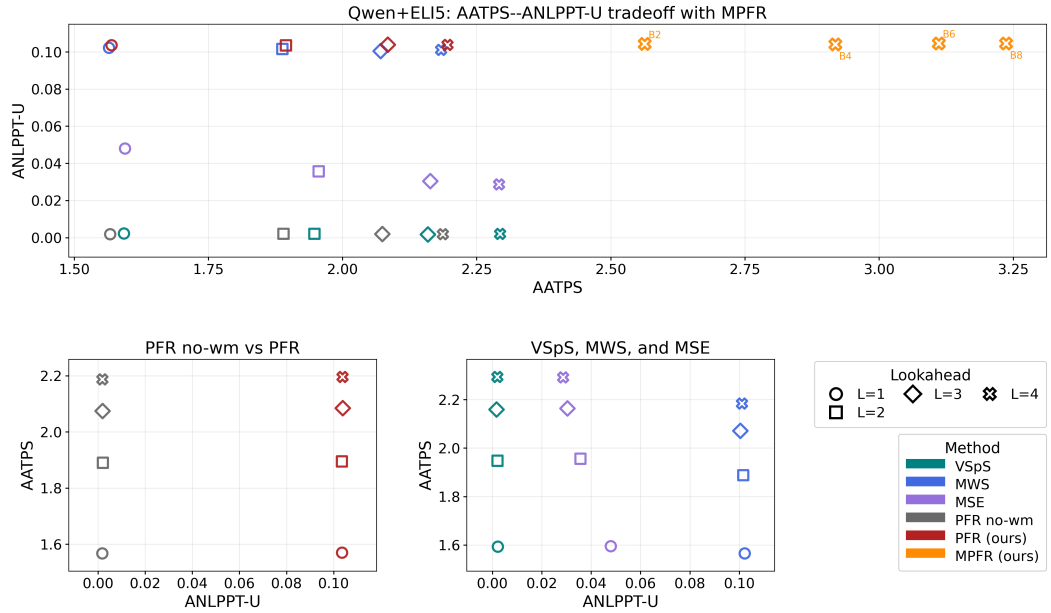


Figure 3: Single-draft (AATPS, ANLPPT-U) frontier on Qwen2.5-7B-Instruct / Qwen2.5-0.5B-Instruct × ELI5. PFR dominates MWS/MSE simultaneously on both axes; attaching the watermark does not decrease AATPS.

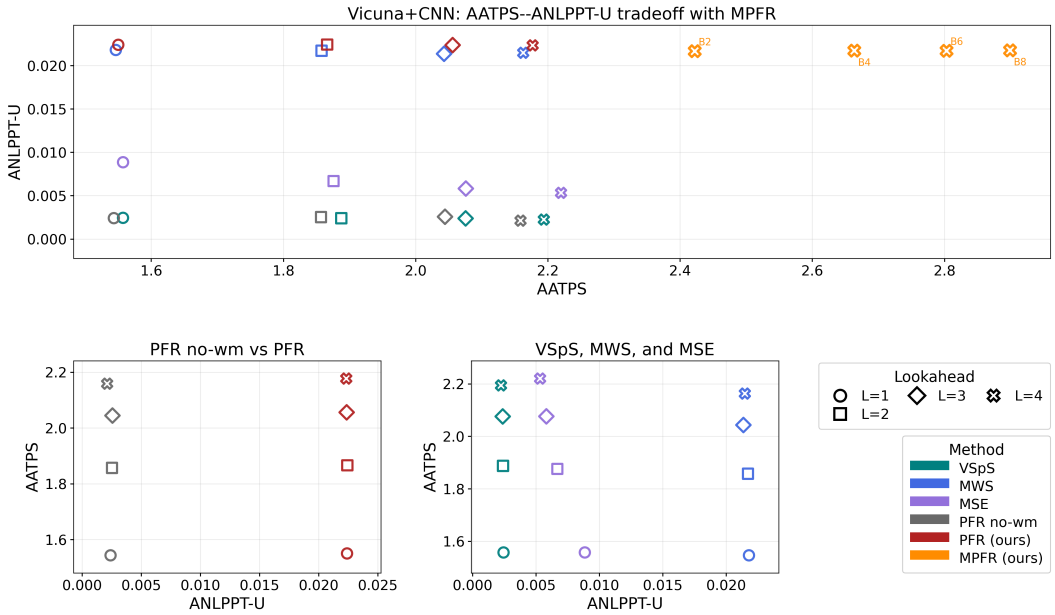


Figure 4: Single-draft (AATPS, ANLPPT-U) frontier on Vicuna-7B-v1.5 / Vicuna-68m × CNN/DAILYMAIL. PFR dominates MWS/MSE simultaneously on both axes; attaching the watermark does not decrease AATPS.

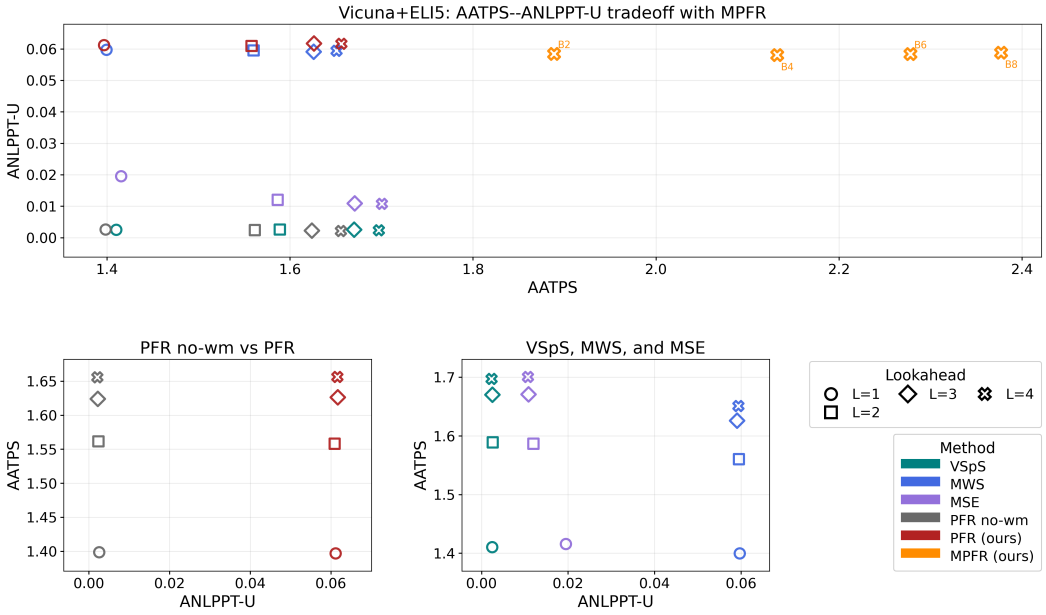


Figure 5: Single-draft (AATPS, ANLPPT-U) frontier on Vicuna-7B-v1.5 / Vicuna-68m × ELI5. PFR dominates MWS/MSE simultaneously on both axes; attaching the watermark does not decrease AATPS.

Table 2: Single-draft results on Qwen2.5-7B-Instruct / Qwen2.5-0.5B-Instruct × CNN/DAILYMAIL. Each cell: mean ± std across 1000 prompts.

L	Method	AATPS	Token rate	ANLPPT-U	ANLPPT-Li	ANLPPT-PL	LPPL
1	BASIC-UWM	1.000 ± 0.000	33.353 ± 2.815	0.048 ± 0.035	0.035 ± 0.029	0.051 ± 0.036	0.473 ± 0.134
	VSPS	1.641 ± 0.070	33.229 ± 1.944	0.003 ± 0.005	0.003 ± 0.006	0.003 ± 0.006	0.477 ± 0.136
	PFR-NOWM	1.623 ± 0.075	32.147 ± 2.161	0.002 ± 0.006	0.003 ± 0.006	0.002 ± 0.006	0.474 ± 0.140
	PFR	1.625 ± 0.072	33.321 ± 1.706	0.051 ± 0.036	0.036 ± 0.030	0.053 ± 0.038	0.470 ± 0.143
	MSE	1.642 ± 0.070	20.578 ± 2.645	0.023 ± 0.021	0.016 ± 0.017	0.025 ± 0.023	0.472 ± 0.136
	MWS	1.621 ± 0.072	20.136 ± 2.952	0.049 ± 0.035	0.034 ± 0.029	0.052 ± 0.036	0.475 ± 0.133
	MSE-PSEUDO	1.641 ± 0.072	20.703 ± 2.420	0.033 ± 0.028	0.023 ± 0.023	0.035 ± 0.029	0.475 ± 0.138
2	BASIC-UWM	1.000 ± 0.000	33.353 ± 2.815	0.048 ± 0.035	0.035 ± 0.029	0.051 ± 0.036	0.473 ± 0.134
	VSPS	2.055 ± 0.147	29.399 ± 2.693	0.002 ± 0.006	0.002 ± 0.006	0.002 ± 0.006	0.482 ± 0.139
	PFR-NOWM	2.014 ± 0.151	30.119 ± 2.438	0.002 ± 0.005	0.003 ± 0.005	0.002 ± 0.005	0.474 ± 0.140
	PFR	2.016 ± 0.154	29.746 ± 2.374	0.051 ± 0.036	0.036 ± 0.030	0.053 ± 0.038	0.471 ± 0.143
	MSE	2.056 ± 0.149	16.954 ± 2.182	0.016 ± 0.017	0.012 ± 0.014	0.018 ± 0.019	0.475 ± 0.139
	MWS	2.012 ± 0.155	17.915 ± 2.679	0.048 ± 0.035	0.034 ± 0.028	0.052 ± 0.036	0.473 ± 0.134
	MSE-PSEUDO	2.053 ± 0.150	17.768 ± 2.415	0.030 ± 0.026	0.022 ± 0.021	0.033 ± 0.028	0.481 ± 0.143
3	BASIC-UWM	1.000 ± 0.000	33.353 ± 2.815	0.048 ± 0.035	0.035 ± 0.029	0.051 ± 0.036	0.473 ± 0.134
	VSPS	2.327 ± 0.228	26.272 ± 2.968	0.002 ± 0.006	0.002 ± 0.006	0.003 ± 0.006	0.481 ± 0.141
	PFR-NOWM	2.270 ± 0.228	26.709 ± 2.682	0.002 ± 0.006	0.002 ± 0.006	0.002 ± 0.005	0.474 ± 0.140
	PFR	2.272 ± 0.225	25.999 ± 2.742	0.051 ± 0.036	0.037 ± 0.030	0.053 ± 0.038	0.472 ± 0.144
	MSE	2.330 ± 0.228	13.453 ± 1.603	0.015 ± 0.017	0.011 ± 0.014	0.017 ± 0.018	0.476 ± 0.140
	MWS	2.262 ± 0.222	16.187 ± 2.509	0.048 ± 0.035	0.034 ± 0.028	0.051 ± 0.036	0.474 ± 0.134
	MSE-PSEUDO	2.337 ± 0.232	14.933 ± 2.118	0.029 ± 0.026	0.021 ± 0.022	0.032 ± 0.027	0.480 ± 0.138
4	BASIC-UWM	1.000 ± 0.000	33.353 ± 2.815	0.048 ± 0.035	0.035 ± 0.029	0.051 ± 0.036	0.473 ± 0.134
	VSPS	2.513 ± 0.288	23.229 ± 3.176	0.002 ± 0.006	0.002 ± 0.005	0.002 ± 0.006	0.475 ± 0.134
	PFR-NOWM	2.439 ± 0.292	23.285 ± 2.840	0.002 ± 0.005	0.002 ± 0.005	0.002 ± 0.006	0.474 ± 0.142
	PFR	2.446 ± 0.286	23.075 ± 2.755	0.050 ± 0.036	0.036 ± 0.030	0.053 ± 0.038	0.471 ± 0.144
	MSE	2.525 ± 0.298	12.623 ± 1.893	0.014 ± 0.016	0.010 ± 0.013	0.016 ± 0.017	0.479 ± 0.145
	MWS	2.432 ± 0.287	13.592 ± 2.213	0.048 ± 0.035	0.034 ± 0.028	0.052 ± 0.036	0.475 ± 0.132
	MSE-PSEUDO	2.499 ± 0.287	13.494 ± 1.934	0.029 ± 0.026	0.021 ± 0.021	0.031 ± 0.027	0.481 ± 0.141

Table 3: Single-draft results on Qwen2.5-7B-Instruct / Qwen2.5-0.5B-Instruct × ELI5. Each cell: mean ± std across 1000 prompts.

L	Method	AATPS	Token rate	ANLPPT-U	ANLPPT-Li	ANLPPT-PL	LPPL
1	BASIC-UWM	1.000 ± 0.000	26.132 ± 1.924	0.100 ± 0.045	0.072 ± 0.039	0.104 ± 0.046	0.696 ± 0.141
	VSPS	1.593 ± 0.060	32.090 ± 1.889	0.002 ± 0.005	0.002 ± 0.005	0.002 ± 0.005	0.701 ± 0.133
	PFR-NOWM	1.567 ± 0.061	33.046 ± 1.405	0.002 ± 0.004	0.002 ± 0.004	0.002 ± 0.004	0.703 ± 0.139
	PFR	1.570 ± 0.062	33.213 ± 1.487	0.104 ± 0.048	0.075 ± 0.039	0.107 ± 0.049	0.699 ± 0.134
	MSE	1.595 ± 0.060	18.218 ± 2.022	0.048 ± 0.030	0.032 ± 0.024	0.053 ± 0.032	0.704 ± 0.135
	MWS	1.565 ± 0.063	20.177 ± 2.688	0.102 ± 0.047	0.075 ± 0.040	0.106 ± 0.048	0.700 ± 0.137
	MSE-PSEUDO	1.593 ± 0.062	19.553 ± 2.107	0.068 ± 0.038	0.049 ± 0.032	0.071 ± 0.039	0.703 ± 0.145
2	BASIC-UWM	1.000 ± 0.000	26.132 ± 1.924	0.100 ± 0.045	0.072 ± 0.039	0.104 ± 0.046	0.696 ± 0.141
	VSPS	1.948 ± 0.116	27.350 ± 2.434	0.002 ± 0.005	0.002 ± 0.005	0.002 ± 0.004	0.701 ± 0.136
	PFR-NOWM	1.890 ± 0.118	28.719 ± 1.782	0.002 ± 0.005	0.002 ± 0.004	0.002 ± 0.004	0.698 ± 0.137
	PFR	1.895 ± 0.125	28.851 ± 1.937	0.104 ± 0.048	0.075 ± 0.039	0.107 ± 0.049	0.699 ± 0.134
	MSE	1.956 ± 0.126	14.867 ± 1.179	0.036 ± 0.024	0.024 ± 0.020	0.040 ± 0.026	0.704 ± 0.140
	MWS	1.888 ± 0.121	16.985 ± 2.208	0.102 ± 0.046	0.074 ± 0.039	0.105 ± 0.047	0.700 ± 0.142
	MSE-PSEUDO	1.951 ± 0.118	24.767 ± 2.016	0.062 ± 0.034	0.043 ± 0.028	0.066 ± 0.036	0.706 ± 0.136
3	BASIC-UWM	1.000 ± 0.000	26.132 ± 1.924	0.100 ± 0.045	0.072 ± 0.039	0.104 ± 0.046	0.696 ± 0.141
	VSPS	2.159 ± 0.175	23.619 ± 2.707	0.002 ± 0.004	0.002 ± 0.004	0.002 ± 0.003	0.706 ± 0.139
	PFR-NOWM	2.074 ± 0.167	24.315 ± 2.065	0.002 ± 0.004	0.002 ± 0.005	0.002 ± 0.004	0.710 ± 0.139
	PFR	2.084 ± 0.174	24.675 ± 2.059	0.104 ± 0.049	0.075 ± 0.041	0.107 ± 0.049	0.697 ± 0.135
	MSE	2.164 ± 0.175	13.359 ± 2.005	0.030 ± 0.023	0.020 ± 0.019	0.035 ± 0.025	0.705 ± 0.135
	MWS	2.071 ± 0.169	14.243 ± 1.924	0.100 ± 0.046	0.073 ± 0.040	0.104 ± 0.047	0.698 ± 0.138
	MSE-PSEUDO	2.169 ± 0.172	21.339 ± 2.086	0.061 ± 0.034	0.043 ± 0.028	0.064 ± 0.035	0.709 ± 0.133
4	BASIC-UWM	1.000 ± 0.000	26.132 ± 1.924	0.100 ± 0.045	0.072 ± 0.039	0.104 ± 0.046	0.696 ± 0.141
	VSPS	2.294 ± 0.220	17.142 ± 3.485	0.002 ± 0.005	0.002 ± 0.005	0.002 ± 0.004	0.705 ± 0.137
	PFR-NOWM	2.187 ± 0.204	21.512 ± 2.145	0.002 ± 0.004	0.002 ± 0.005	0.002 ± 0.004	0.703 ± 0.138
	PFR	2.196 ± 0.214	21.319 ± 2.212	0.104 ± 0.048	0.075 ± 0.041	0.107 ± 0.049	0.698 ± 0.134
	MSE	2.292 ± 0.220	13.200 ± 2.021	0.029 ± 0.021	0.019 ± 0.017	0.033 ± 0.024	0.701 ± 0.136
	MWS	2.184 ± 0.208	11.570 ± 1.496	0.101 ± 0.046	0.073 ± 0.039	0.105 ± 0.047	0.696 ± 0.141
	MSE-PSEUDO	2.302 ± 0.214	18.298 ± 1.942	0.060 ± 0.033	0.042 ± 0.028	0.063 ± 0.035	0.705 ± 0.132

Table 4: Single-draft results on Vicuna-7B-v1.5 / Vicuna-68m × CNN/DAILYMAIL. Each cell: mean ± std across 1000 prompts.

L	Method	AATPS	Token rate	ANLPPT-U	ANLPPT-Li	ANLPPT-PL	LPPL
1	BASIC-UWM	1.000 ± 0.000	37.058 ± 0.848	0.021 ± 0.024	0.016 ± 0.019	0.022 ± 0.025	0.274 ± 0.190
	VSPS	1.557 ± 0.086	48.890 ± 3.439	0.002 ± 0.006	0.003 ± 0.006	0.002 ± 0.006	0.275 ± 0.167
	PFR-NOWM	1.544 ± 0.088	37.027 ± 5.064	0.002 ± 0.006	0.003 ± 0.006	0.002 ± 0.005	0.269 ± 0.121
	PFR	1.550 ± 0.090	38.256 ± 6.092	0.022 ± 0.024	0.017 ± 0.019	0.023 ± 0.025	0.272 ± 0.128
	MSE	1.557 ± 0.088	45.513 ± 4.490	0.009 ± 0.013	0.007 ± 0.011	0.010 ± 0.015	0.269 ± 0.123
	MWS	1.547 ± 0.091	30.814 ± 4.078	0.022 ± 0.024	0.017 ± 0.020	0.023 ± 0.026	0.276 ± 0.192
	MSE-PSEUDO	1.555 ± 0.088	27.597 ± 2.405	0.015 ± 0.018	0.012 ± 0.016	0.016 ± 0.018	0.279 ± 0.118
2	BASIC-UWM	1.000 ± 0.000	37.058 ± 0.848	0.021 ± 0.024	0.016 ± 0.019	0.022 ± 0.025	0.274 ± 0.190
	VSPS	1.888 ± 0.179	53.865 ± 5.770	0.002 ± 0.006	0.002 ± 0.006	0.002 ± 0.006	0.268 ± 0.129
	PFR-NOWM	1.857 ± 0.178	40.154 ± 5.953	0.003 ± 0.005	0.003 ± 0.006	0.003 ± 0.005	0.273 ± 0.123
	PFR	1.866 ± 0.183	42.639 ± 7.272	0.022 ± 0.024	0.017 ± 0.019	0.023 ± 0.025	0.272 ± 0.127
	MSE	1.876 ± 0.186	46.276 ± 7.794	0.007 ± 0.010	0.006 ± 0.009	0.007 ± 0.011	0.268 ± 0.131
	MWS	1.858 ± 0.181	31.588 ± 5.453	0.022 ± 0.024	0.016 ± 0.020	0.023 ± 0.025	0.278 ± 0.193
	MSE-PSEUDO	1.877 ± 0.182	29.023 ± 4.118	0.015 ± 0.018	0.012 ± 0.015	0.015 ± 0.020	0.273 ± 0.127
3	BASIC-UWM	1.000 ± 0.000	37.058 ± 0.848	0.021 ± 0.024	0.016 ± 0.019	0.022 ± 0.025	0.274 ± 0.190
	VSPS	2.076 ± 0.254	53.835 ± 7.463	0.002 ± 0.006	0.003 ± 0.006	0.002 ± 0.005	0.276 ± 0.212
	PFR-NOWM	2.044 ± 0.259	42.513 ± 7.326	0.003 ± 0.006	0.003 ± 0.006	0.002 ± 0.006	0.274 ± 0.128
	PFR	2.056 ± 0.263	41.813 ± 7.586	0.022 ± 0.024	0.017 ± 0.019	0.023 ± 0.025	0.272 ± 0.127
	MSE	2.076 ± 0.260	38.713 ± 8.338	0.006 ± 0.009	0.005 ± 0.008	0.006 ± 0.010	0.266 ± 0.118
	MWS	2.043 ± 0.258	28.960 ± 4.759	0.021 ± 0.024	0.016 ± 0.020	0.022 ± 0.026	0.277 ± 0.191
	MSE-PSEUDO	2.062 ± 0.257	32.256 ± 6.945	0.015 ± 0.019	0.012 ± 0.016	0.015 ± 0.020	0.277 ± 0.125
4	BASIC-UWM	1.000 ± 0.000	37.058 ± 0.848	0.021 ± 0.024	0.016 ± 0.019	0.022 ± 0.025	0.274 ± 0.190
	VSPS	2.194 ± 0.324	53.940 ± 8.237	0.002 ± 0.005	0.002 ± 0.005	0.002 ± 0.005	0.271 ± 0.120
	PFR-NOWM	2.159 ± 0.326	40.083 ± 7.877	0.002 ± 0.005	0.002 ± 0.005	0.002 ± 0.005	0.268 ± 0.117
	PFR	2.177 ± 0.323	41.800 ± 9.355	0.022 ± 0.023	0.017 ± 0.019	0.023 ± 0.025	0.272 ± 0.127
	MSE	2.220 ± 0.330	32.730 ± 6.784	0.005 ± 0.008	0.005 ± 0.008	0.006 ± 0.009	0.264 ± 0.119
	MWS	2.163 ± 0.319	27.253 ± 4.410	0.021 ± 0.024	0.016 ± 0.019	0.023 ± 0.025	0.277 ± 0.191
	MSE-PSEUDO	2.183 ± 0.317	33.007 ± 6.806	0.014 ± 0.018	0.011 ± 0.015	0.014 ± 0.018	0.277 ± 0.121

Table 5: Single-draft results on Vicuna-7B-v1.5 / Vicuna-68m × ELI5. Each cell: $\mathrm{mean} \pm \mathrm{std}$ across 1000 prompts.

L	Method	AATPS	Token rate	ANLPPT-U	ANLPPT-Li	ANLPPT-PL	LPPL
1	BASIC-UWM	1.000 ± 0.000	38.536 ± 1.896	0.060 ± 0.040	0.044 ± 0.033	0.062 ± 0.042	0.507 ± 0.152
	VSPS	1.410 ± 0.063	46.306 ± 3.178	0.002 ± 0.005	0.003 ± 0.005	0.002 ± 0.005	0.511 ± 0.150
	PFR-NOWM	1.399 ± 0.063	33.636 ± 4.757	0.003 ± 0.006	0.003 ± 0.007	0.002 ± 0.005	0.509 ± 0.152
	PFR	1.397 ± 0.065	35.231 ± 5.397	0.061 ± 0.043	0.045 ± 0.038	0.063 ± 0.042	0.513 ± 0.162
	MSE	1.415 ± 0.062	44.719 ± 3.192	0.020 ± 0.020	0.014 ± 0.017	0.022 ± 0.022	0.509 ± 0.147
	MWS	1.399 ± 0.064	26.309 ± 3.816	0.060 ± 0.041	0.044 ± 0.034	0.062 ± 0.042	0.507 ± 0.151
	MSE-PSEUDO	1.416 ± 0.063	28.987 ± 4.863	0.040 ± 0.030	0.029 ± 0.025	0.042 ± 0.031	0.504 ± 0.152
2	BASIC-UWM	1.000 ± 0.000	38.536 ± 1.896	0.060 ± 0.040	0.044 ± 0.033	0.062 ± 0.042	0.507 ± 0.152
	VSPS	1.589 ± 0.104	47.569 ± 3.837	0.003 ± 0.006	0.003 ± 0.006	0.003 ± 0.006	0.509 ± 0.146
	PFR-NOWM	1.562 ± 0.103	34.678 ± 4.452	0.002 ± 0.005	0.002 ± 0.005	0.002 ± 0.006	0.505 ± 0.147
	PFR	1.558 ± 0.106	34.425 ± 4.996	0.061 ± 0.043	0.044 ± 0.038	0.063 ± 0.042	0.513 ± 0.162
	MSE	1.587 ± 0.105	33.648 ± 4.994	0.012 ± 0.014	0.009 ± 0.012	0.014 ± 0.016	0.509 ± 0.147
	MWS	1.560 ± 0.109	23.481 ± 2.014	0.060 ± 0.040	0.043 ± 0.034	0.062 ± 0.042	0.510 ± 0.157
	MSE-PSEUDO	1.593 ± 0.110	28.595 ± 4.733	0.040 ± 0.032	0.029 ± 0.027	0.041 ± 0.032	0.509 ± 0.152
3	BASIC-UWM	1.000 ± 0.000	38.536 ± 1.896	0.060 ± 0.040	0.044 ± 0.033	0.062 ± 0.042	0.507 ± 0.152
	VSPS	1.670 ± 0.134	45.003 ± 5.451	0.003 ± 0.006	0.002 ± 0.005	0.002 ± 0.006	0.513 ± 0.157
	PFR-NOWM	1.624 ± 0.132	31.214 ± 4.662	0.002 ± 0.005	0.002 ± 0.005	0.002 ± 0.005	0.506 ± 0.149
	PFR	1.626 ± 0.130	35.694 ± 5.835	0.062 ± 0.043	0.045 ± 0.038	0.063 ± 0.042	0.515 ± 0.162
	MSE	1.671 ± 0.136	26.134 ± 3.956	0.011 ± 0.014	0.008 ± 0.012	0.012 ± 0.015	0.499 ± 0.145
	MWS	1.626 ± 0.136	22.536 ± 2.127	0.059 ± 0.040	0.043 ± 0.033	0.061 ± 0.041	0.508 ± 0.152
	MSE-PSEUDO	1.666 ± 0.134	28.611 ± 4.748	0.037 ± 0.028	0.027 ± 0.023	0.039 ± 0.029	0.505 ± 0.148
4	BASIC-UWM	1.000 ± 0.000	38.536 ± 1.896	0.060 ± 0.040	0.044 ± 0.033	0.062 ± 0.042	0.507 ± 0.152
	VSPS	1.697 ± 0.152	43.287 ± 5.333	0.002 ± 0.005	0.002 ± 0.006	0.002 ± 0.005	0.510 ± 0.152
	PFR-NOWM	1.656 ± 0.145	28.914 ± 4.147	0.002 ± 0.005	0.002 ± 0.005	0.002 ± 0.005	0.503 ± 0.149
	PFR	1.656 ± 0.146	32.316 ± 5.921	0.062 ± 0.043	0.045 ± 0.038	0.063 ± 0.042	0.515 ± 0.162
	MSE	1.700 ± 0.154	23.993 ± 3.469	0.011 ± 0.014	0.008 ± 0.012	0.012 ± 0.015	0.514 ± 0.150
	MWS	1.651 ± 0.149	22.198 ± 3.679	0.059 ± 0.039	0.043 ± 0.033	0.062 ± 0.041	0.509 ± 0.153
	MSE-PSEUDO	1.703 ± 0.155	26.004 ± 4.603	0.037 ± 0.030	0.027 ± 0.024	0.039 ± 0.031	0.508 ± 0.147

evaluation to the 2×2 matrix {Qwen2.5-7B-Instruct, Vicuna-7B-v1.5} × {CNN/DAILYMAIL, ELI5}, using the target/drafter pairs `Qwen2.5-7B-Instruct`/`Qwen2.5-0.5B-Instruct` and `lmsys/vicuna-7b-v1.5`/`double7/vicuna-68m`. Each cell uses n=1000 prompts, lookahead L=4, and draft counts $B \in \{2, 4, 6, 8\}$, with the sampling parameters of Appendix D.1.

Decoders. MPFR is our keyed multi-draft Poisson sampler; INVARIANT is the unwatermarked *InvariantMultiDraftStrategy* of Rowan et al. (2025) run in the same harness on the same prompt set. The two decoders share the draft/verify mechanism and differ only in whether the per-context watermark key is applied.

Per-cell tables. Tables 10–13 report all quality and watermark-strength metrics per (decoder, B). Columns: AATPS, token rate in tokens/s (TR), ANLPPT-{U, Li, PL}, and LPPL. The corresponding TPR-vs-T_{eval} curves are reported in Figure 6 of Appendix D.5.

Table 6: Multi-draft results on Qwen2.5-7B-Instruct × CNN/DAILYMAIL, L=4.

Decoder	B	AATPS	TR	ANLPPT-U	ANLPPT-Li	ANLPPT-PL	LPPL
MPFR	2	2.797 ± 0.001	25.11 ± 0.22	0.051 ± 0.001	0.036 ± 0.001	0.053 ± 0.001	0.474 ± 0.002
MPFR	4	3.123 ± 0.002	27.50 ± 0.31	0.051 ± 0.001	0.036 ± 0.001	0.053 ± 0.001	0.474 ± 0.002
MPFR	6	3.294 ± 0.003	28.50 ± 0.49	0.050 ± 0.001	0.036 ± 0.001	0.053 ± 0.001	0.474 ± 0.001
MPFR	8	3.409 ± 0.002	29.13 ± 0.59	0.051 ± 0.002	0.036 ± 0.001	0.053 ± 0.002	0.474 ± 0.002
INVARIANT	2	2.803 ± 0.007	24.78 ± 0.14	0.002 ± 0.000	0.003 ± 0.000	0.002 ± 0.000	0.483 ± 0.005
INVARIANT	4	3.109 ± 0.008	27.12 ± 0.20	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.484 ± 0.003
INVARIANT	6	3.272 ± 0.010	27.84 ± 0.60	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.481 ± 0.000
INVARIANT	8	3.384 ± 0.012	28.62 ± 0.54	0.002 ± 0.000	0.003 ± 0.000	0.002 ± 0.000	0.476 ± 0.004

Speculative-acceptance parity. Within each cell, MPFR remains close to INVARIANT in AATPS across $B \in \{2, 4, 6, 8\}$, with small deviations in both directions depending on the model–dataset cell. This indicates that the target-side keyed-race construction does not introduce a systematic per-step acceptance penalty. Token rate is reported as a hardware- and implementation-dependent diagnostic,

Table 7: Multi-draft results on Qwen2.5-7B-Instruct × ELI5, L=4.

Decoder	B	AATPS	TR	ANLPPT-U	ANLPPT-Li	ANLPPT-PL	LPPL
MPFR	2	2.563 ± 0.007	23.57 ± 0.15	0.104 ± 0.001	0.075 ± 0.001	0.108 ± 0.001	0.705 ± 0.001
MPFR	4	2.918 ± 0.008	26.46 ± 0.25	0.104 ± 0.001	0.076 ± 0.001	0.108 ± 0.001	0.706 ± 0.000
MPFR	6	3.109 ± 0.005	27.81 ± 0.35	0.103 ± 0.001	0.075 ± 0.001	0.107 ± 0.001	0.705 ± 0.000
MPFR	8	3.235 ± 0.010	28.73 ± 0.49	0.104 ± 0.002	0.075 ± 0.002	0.108 ± 0.001	0.706 ± 0.001
INVARIANT	2	2.523 ± 0.004	22.73 ± 0.22	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.747 ± 0.005
INVARIANT	4	2.850 ± 0.010	25.60 ± 0.26	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.736 ± 0.002
INVARIANT	6	3.039 ± 0.004	27.01 ± 0.32	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.728 ± 0.002
INVARIANT	8	3.156 ± 0.009	28.09 ± 0.36	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.728 ± 0.002

Table 8: Multi-draft results on Llama-3.1-8B-Instruct × CNN/DAILYMAIL, $L = 4$. Mean ± std over 3 random seeds, 1000 prompts each. The drafter is Llama-3.2-1B-Instruct, the official distillation of the target sharing the same tokenizer.

Decoder	B	AATPS	TR	ANLPPT-U	ANLPPT-Li	ANLPPT-PL	LPPL
MPFR	2	3.459 ± 0.015	37.44 ± 0.57	0.058 ± 0.001	0.041 ± 0.000	0.061 ± 0.001	0.514 ± 0.002
MPFR	4	3.784 ± 0.013	40.20 ± 0.28	0.058 ± 0.000	0.041 ± 0.000	0.062 ± 0.001	0.514 ± 0.003
MPFR	6	3.935 ± 0.005	41.71 ± 0.30	0.058 ± 0.000	0.041 ± 0.000	0.061 ± 0.001	0.514 ± 0.003
MPFR	8	4.034 ± 0.005	42.04 ± 0.81	0.058 ± 0.001	0.041 ± 0.000	0.061 ± 0.001	0.514 ± 0.003
INVARIANT	2	3.431 ± 0.005	36.79 ± 0.37	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.514 ± 0.003
INVARIANT	4	3.741 ± 0.007	39.31 ± 0.29	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.508 ± 0.001
INVARIANT	6	3.894 ± 0.007	40.51 ± 0.25	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.507 ± 0.001
INVARIANT	8	3.987 ± 0.010	39.87 ± 0.66	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.505 ± 0.003

since it also reflects batching, hashing, and watermark-bookkeeping overhead beyond acceptance behavior.

Stability of the watermark signal across B. ANLPPT-U, ANLPPT-Li, and ANLPPT-PL for MPFR remain within ± 0.001 across $B \in \{2, 4, 6, 8\}$ on every cell of Tables 10–13. The per-token detection signal therefore does not dilute with the number of drafts, in contrast to naive multi-draft watermarking, which would re-introduce the efficiency–detectability trade-off of Hu and Huang (2024) at the multi-draft level.

Distortion-free audit. LPPL of MPFR outputs falls within ± 0.013 nats/token of INVARIANT on every cell. This empirical audit is consistent with the exact marginal-preservation guarantee of the MPFR construction and indicates that the watermark does not introduce a measurable shift in target-model log-perplexity in the multi-draft regime.

D.5 DRAFTER-SUBSTITUTION ABLATION: FULL RESULTS

Drafter conditions. D_0: Qwen2.5-0.5B-Instruct at T=1.0 (default). D_1: Qwen2.5-1.5B-Instruct at T=1.0 (3× scale, model swap). D_2: Qwen2.5-0.5B-Instruct at T=0.5 (sharper drafter). D_3: Qwen2.5-0.5B-Instruct at T=1.5 (more diffuse drafter). The target temperature, watermark key, and all other parameters are held fixed across D_0–D_3.

Detection statistic. For the Aaronson Gamma-tail test, the per-token score is $s_t = -\log(1 - U_t)$, where U_t is the per-token uniform recovered from the watermark code. Under the no-watermark null, the cumulative score $S_T = \sum_{t \le T} s_t$ is $\mathrm{Gamma}(T, 1)$; we declare detection at FPR= α when $\Pr_{H_0}[\mathrm{Gamma}(T, 1) > S_T] \le \alpha$, computed via the regularised upper incomplete gamma. We report α=1% throughout.

Full detection across models and datasets. Figure 6 extends the default-drafter detection comparison of Figure 2(a) to all four (TARGET, DATASET) cells of the main 2×2 result. In every cell, PFR/MPFR (ours) coincide with the strong-watermark ceiling defined by autoregressive BASIC-UWM and MWS, while MSE and MSE-PSEUDO (He et al., 2026) are uniformly weaker. The size of the gap is dataset-dependent — largest on CNN/DAILYMAIL, smallest on the longer-form ELI5 where every scheme approaches saturation by T_{eval}=128 — but the ranking is invariant. This

Table 9: Multi-draft results on Llama-3.1-8B-Instruct × ELI5, $L = 4$. Mean ± std over 3 random seeds, 1000 prompts each.

Decoder	B	AATPS	TR	ANLPPT-U	ANLPPT-Li	ANLPPT-PL	LPPL
MPFR	2	3.103 ± 0.006	34.29 ± 0.35	0.135 ± 0.001	0.099 ± 0.001	0.139 ± 0.001	0.811 ± 0.004
MPFR	4	3.455 ± 0.010	37.79 ± 0.38	0.135 ± 0.002	0.099 ± 0.001	0.139 ± 0.002	0.812 ± 0.003
MPFR	6	3.635 ± 0.001	39.59 ± 0.43	0.135 ± 0.002	0.099 ± 0.001	0.139 ± 0.002	0.812 ± 0.005
MPFR	8	3.749 ± 0.002	39.76 ± 0.80	0.135 ± 0.001	0.098 ± 0.000	0.139 ± 0.001	0.811 ± 0.004
INVARIANT	2	3.080 ± 0.011	33.72 ± 0.60	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.803 ± 0.003
INVARIANT	4	3.408 ± 0.006	37.11 ± 0.12	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.794 ± 0.004
INVARIANT	6	3.588 ± 0.008	39.12 ± 0.31	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.795 ± 0.008
INVARIANT	8	3.700 ± 0.008	39.37 ± 0.58	0.002 ± 0.000	0.002 ± 0.000	0.002 ± 0.000	0.790 ± 0.003

Table 10: Multi-draft results on Qwen2.5-7B-Instruct × CNN/DAILYMAIL, L=4.

Decoder	B	AATPS	TR	ANLPPT-U	ANLPPT-Li	ANLPPT-PL	LPPL
MPFR	2	2.799	23.18	0.052	0.037	0.055	0.475
MPFR	4	3.124	24.89	0.052	0.037	0.055	0.474
MPFR	6	3.298	25.39	0.052	0.037	0.055	0.476
MPFR	8	3.410	25.17	0.053	0.037	0.055	0.476
INVARIANT	2	2.804	25.03	0.003	0.003	0.002	0.487
INVARIANT	4	3.114	26.63	0.002	0.002	0.002	0.481
INVARIANT	6	3.275	26.79	0.002	0.003	0.002	0.482
INVARIANT	8	3.387	26.38	0.002	0.003	0.002	0.479

confirms that the detection-strength conclusion drawn from the anchor cell in Figure 2(a) is not specific to a single (TARGET, DATASET) pair.

Pairwise ROUGE-L under drafter substitution. For each prompt p and decoder d we have one realised output token sequence per drafter condition. We compute ROUGE-L F1 over each pair (D_i, D_j) of drafter conditions on these token sequences, then report the per-prompt minimum over the six pairs (worst pair) averaged across prompts (Table 15). An unbiased verify rule that depends only on the target logits and the watermark key produces identical sequences across all drafters, so this quantity should approach 1.

E MORE ON WATERMARK

In this section, we discuss different watermarks that are related to our scheme.

E.1 GUMBEL WATERMARK

We use a common notation for the scored positions throughout this subsection. Let $\mathcal{T} := \{m + 1, \ldots, n\}$ and $M := |\mathcal{T}| = n - m$. For each $t \in \mathcal{T}$, let $r_t : \mathcal{V} \to (0, 1)$ be the keyed random function, and define

$$U_t := r_t(w_t), \qquad A_t := -\log(1 - U_t).$$

Aaronson's Gumbel watermark score (Aaronson and Kirchner, 2023) is

$$S_{\mathrm{A}} := \sum_{t \in \mathcal{T}} A_t = \sum_{t \in \mathcal{T}} -\log(1 - r_t(w_t)).$$

Under the null hypothesis, i.e., when $y_{1:n}$ is not generated from the watermarked model, the variables $(U_t)_{t \in \mathcal{T}}$ are independent $\mathrm{Unif}(0, 1)$ random variables. Hence $(A_t)_{t \in \mathcal{T}}$ are independent $\mathrm{Exp}(1)$ random variables, and

$$\mathbf{E}\left[S_{\mathrm{A}}\right] = M.$$

If $y_{1:n}$ is generated from the Gumbel-watermarked model, then

$$\mathbf{E}\left[S_{\mathrm{A}}\right] \geq M + \left(\frac{\pi^2}{6} - 1\right) \sum_{t \in \mathcal{T}} \mathbf{E}\left[\mathrm{Entropy}\left[p_t(\cdot)\right]\right],$$

Table 11: Multi-draft results on Qwen2.5-7B-Instruct × ELI5, L=4.

Decoder	B	AATPS	TR	ANLPPT-U	ANLPPT-Li	ANLPPT-PL	LPPL
MPFR	2	2.563	22.38	0.104	0.076	0.108	0.704
MPFR	4	2.919	24.48	0.104	0.076	0.107	0.703
MPFR	6	3.111	25.54	0.105	0.076	0.108	0.706
MPFR	8	3.237	25.83	0.105	0.076	0.108	0.704
INVARIANT	2	2.519	23.18	0.002	0.002	0.002	0.750
INVARIANT	4	2.852	25.75	0.002	0.002	0.002	0.733
INVARIANT	6	3.043	27.11	0.002	0.002	0.002	0.730
INVARIANT	8	3.158	27.96	0.002	0.002	0.002	0.725

Table 12: Multi-draft results on Vicuna-7B-v1.5 × CNN/DAILYMAIL, L=4.

Decoder	B	AATPS	TR	ANLPPT-U	ANLPPT-Li	ANLPPT-PL	LPPL
MPFR	2	2.422	52.10	0.022	0.017	0.022	0.270
MPFR	4	2.664	48.28	0.022	0.017	0.023	0.271
MPFR	6	2.803	44.44	0.022	0.017	0.023	0.270
MPFR	8	2.899	40.76	0.022	0.017	0.023	0.270
INVARIANT	2	2.478	58.15	0.002	0.002	0.002	0.287
INVARIANT	4	2.690	53.12	0.002	0.002	0.002	0.286
INVARIANT	6	2.831	48.41	0.002	0.003	0.002	0.283
INVARIANT	8	2.925	44.01	0.002	0.002	0.002	0.288

Table 13: Multi-draft results on Vicuna-7B-v1.5 × ELI5, L=4.

Decoder	B	AATPS	TR	ANLPPT-U	ANLPPT-Li	ANLPPT-PL	LPPL
MPFR	2	1.889	45.89	0.058	0.043	0.061	0.504
MPFR	4	2.133	47.43	0.058	0.042	0.061	0.504
MPFR	6	2.278	47.42	0.058	0.042	0.062	0.503
MPFR	8	2.377	46.42	0.059	0.043	0.062	0.504
INVARIANT	2	1.884	50.29	0.002	0.002	0.002	0.555
INVARIANT	4	2.108	53.42	0.002	0.002	0.002	0.552
INVARIANT	6	2.248	55.08	0.002	0.003	0.002	0.546
INVARIANT	8	2.361	56.00	0.003	0.003	0.002	0.539

where $p_t = \mathrm{Softmax}(u_t/T)$ and the entropy is measured in nats.

Since $S_{\mathrm{A}} \sim \mathrm{Gamma}(M, 1)$ under the null hypothesis, the corresponding p-value is

$$p_{\mathrm{A}} = \mathbf{P}\big(\mathrm{Gamma}(M,1) \geq S_{\mathrm{A}}\big) = \frac{\Gamma(M, S_{\mathrm{A}})}{\Gamma(M)},$$

where $\Gamma(M)$ is the gamma function and $\Gamma(M, S_{\mathrm{A}})$ is the upper incomplete gamma function. The Average Negative Log P-value Per Token (ANLPPT) for Aaronson's score is

$$\mathrm{ANLPPT}_{\mathrm{A}} := -\frac{1}{M} \log p_{\mathrm{A}}.$$

Following the U-score framework (Hu and Huang, 2024), one may instead aggregate the keyed uniforms directly:

$$S_{\mathrm{U}} := \sum_{t \in \mathcal{T}} U_t.$$

Under the null hypothesis, $U_t \sim \mathrm{Unif}(0, 1)$. Since for $U \sim \mathrm{Unif}(0, 1)$,

$$\mathbf{E}\big[e^{\lambda U}\big] = \frac{e^{\lambda} - 1}{\lambda}, \qquad \lambda \geq 0,$$

we have

$$\mathbf{E}\big[e^{\lambda S_{\mathrm{U}}}\big] = \left(\frac{e^{\lambda} - 1}{\lambda}\right)^{M}.$$

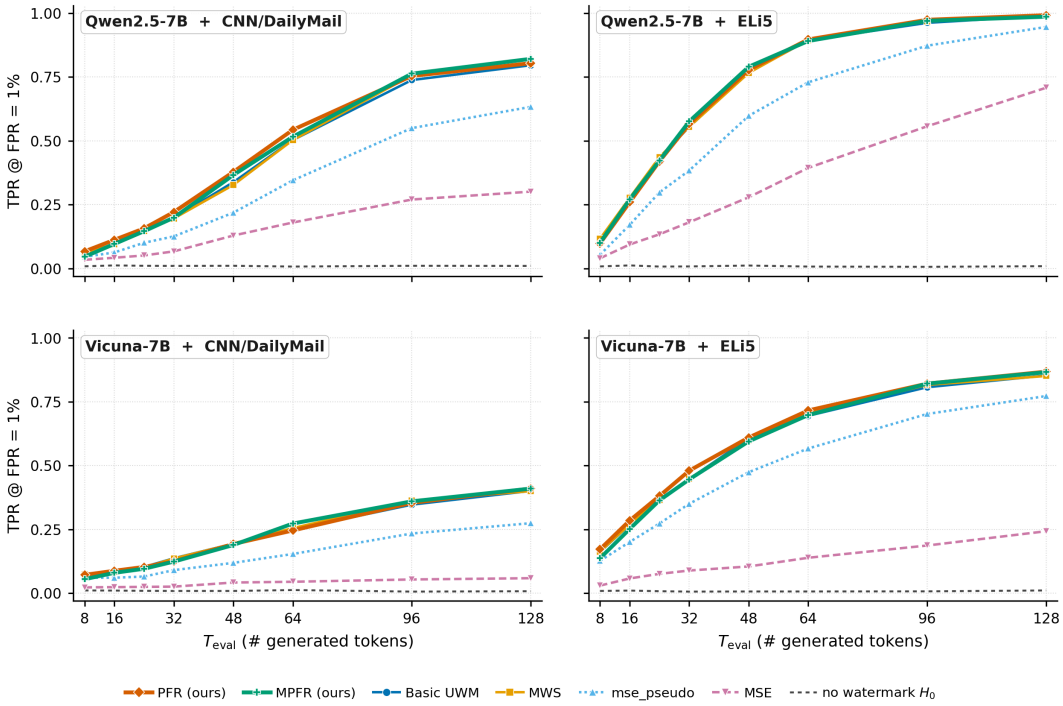


Figure 6: **Full detection across** (TARGET, DATASET) **cells.** TPR@1%FPR (Aaronson Gamma-tail) versus detection token budget T_{eval} at the default drafter D_0, for the seven schemes plotted in Figure 2(a). Each panel is one cell of the main 2×2 result: target ∈ {Qwen2.5-7B-Instruct, Vicuna-7B-v1.5} × dataset ∈ {CNN/DAILYMAIL, ELI5}. PFR/MPFR (ours) match the strong-watermark ceiling (BASIC-UWM, MWS) in every cell, while MSE and MSE-PSEUDO are uniformly weaker. The empirical H_0 (no-watermark) floor sits near the nominal 1% FPR, calibrating the test.

Table 14: **Output quality.** LPPL↓ under target. ROUGE-L↑ vs. gold `highlights`. AATPS↑ accepted target tokens per step.

	Single-draft baselines							*Multi-draft (B=4)*	
Metric	VSPS	PFR-NOWM	BASIC-UWM	MSE	MWS	MSE-PSEUDO	**PFR**	INVARIANT	**MPFR**
LPPL↓	0.474	0.469	0.472	0.470	0.475	0.477	**0.465**	0.484	**0.474**
ROUGE-L↑	0.219	0.222	0.223	0.221	0.223	0.219	**0.223**	0.222	**0.222**
AATPS↑	2.51	2.46	1.00	2.53	2.43	2.52	**2.46**	3.12	**3.13**

Chernoff's bound gives

$$p_{\text{U}} \leq \inf_{\lambda \geq 0} \exp\left(M \log \frac{e^{\lambda} - 1}{\lambda} - \lambda S_{\text{U}} \right),$$

and the corresponding ANLPPT is

$$\text{ANLPPT}_{\text{U}} := -\frac{1}{M} \log p_{\text{U}}.$$

E.2 CONNECTION TO DELTAGUMBEL REWEIGHTING

We now relate the above notation to the DeltaGumbel reweighting rule used in (Hu and Huang, 2024). For each scored position $t \in \mathcal{T}$, let

$$P_t := P(\cdot \mid w_{:t-1})$$

be the target next-token distribution. Define

$$g_t(v) := -\log\big(-\log r_t(v)\big), \qquad v \in \mathcal{V}.$$

Table 15: **Output stability under drafter substitution.** Pairwise ROUGE-L F1 between the realised token sequences across each (D_i, D_j) pair, per-prompt minimum averaged over prompts. An unbiased verify rule produces identical sequences across all drafters, so ROUGE-L $\to 1$.

Decoder	min-pair mean ROUGE-L
BASIC-UWM (no drafter, vacuous bound)	1.000
PFR (ours)	**0.990**
MWS	0.922
PFR-NOWM (no wm)	0.436
MSE-PSEUDO	0.426
MSE	0.406
VSPS (no wm)	0.394

Then $(g_t(v))_{v\in\mathcal{V}}$ are independent standard Gumbel random variables. The DeltaGumbel rule selects

$$a_t^\star := \arg\max_{v\in\mathcal{V}} \{\log P_t(v) + g_t(v)\}.$$

Equivalently, for a fixed key, the reweighted distribution at position t is the point mass

$$Q_t := R_{g_t}(P_t) = \delta_{a_t^\star}.$$

The likelihood-agnostic U-score of (Hu and Huang, 2024) is

$$U_t := \exp\big(-\exp(-g_t(w_t))\big).$$

By the definition of g_t, this is exactly

$$U_t = r_t(w_t).$$

Therefore, Aaronson's per-token score is a deterministic transform of the DeltaGumbel U-score:

$$A_t = -\log(1 - r_t(w_t)) = -\log(1 - U_t).$$

Thus

$$S_{\mathrm{A}} = \sum_{t\in\mathcal{T}} A_t = \sum_{t\in\mathcal{T}} -\log(1 - U_t).$$

In particular, if the full vector $(U_t)_{t\in\mathcal{T}}$ is retained, then Aaronson's score can be computed exactly from the U-scores, and conversely

$$U_t = 1 - \exp(-A_t).$$

However, the aggregate statistics

$$\sum_{t\in\mathcal{T}} U_t \qquad \text{and} \qquad \sum_{t\in\mathcal{T}} -\log(1 - U_t)$$

are not equivalent in general, since they apply different nonlinear transformations before aggregation.

E.3 WATERMARK SCORE BY (LI ET AL., 2025)

The score of Li et al. (2025) is also computed from the same pivot $r_t(w_t)$. For a parameter Δ, define

$$k_\Delta := \left\lfloor \frac{1}{1-\Delta} \right\rfloor, \qquad q_\Delta := 1 - k_\Delta(1-\Delta).$$

Their per-token score is

$$h^\star_{\mathrm{gum},\Delta}(r) = \log\left(k_\Delta r^{\frac{\Delta}{1-\Delta}} + \mathbb{1}\{q_\Delta > 0\} r^{\frac{1-q_\Delta}{q_\Delta}}\right),$$

and the corresponding statistic is

$$S_{\mathrm{Li},\Delta} := \sum_{t\in\mathcal{T}} h^\star_{\mathrm{gum},\Delta}(r_t(w_t)).$$

We now discuss another watermark score for Aaronson's Gumbel Watermark, proposed by (Li et al., 2025), which is a statistically optimal score function for the Gumbel watermark under a class-dependent efficiency criterion. We introduce their result using our notation as follows. Their Gumbel pivot is exactly

$$Y_t^{\mathrm{gum}} := r_t(w_t).$$

Under the null hypothesis,

$$Y_t^{\mathrm{gum}} \sim \mathrm{Unif}[0,1].$$

Under the Gumbel-watermarked alternative, conditional on

$$P_t := P(\cdot|w_{:t-1}),$$

the pivot satisfies

$$\mathbf{P}_{H_1}(Y_t^{\mathrm{gum}} \le r|P_t) = \sum_{v\in\mathcal{V}} P_t(v) r^{1/P_t(v)}, \qquad r \in [0,1],$$

where terms with $P_t(v) = 0$ are omitted.

Aaronson's original score corresponds to the per-token score function

$$h_{\mathrm{A}}(r) := -\log(1-r),$$

so that

$$S_{\mathrm{A}} = \sum_{t\in\mathcal{T}} h_{\mathrm{A}}(r_t(w_t)).$$

In contrast, Li et al. (Li et al., 2025) consider the distribution class

$$\mathcal{P}_\Delta := \left\{ P : \max_{v\in\mathcal{V}} P(v) \le 1-\Delta \right\}$$

and derive the optimal Gumbel score

$$h^\star_{\mathrm{gum},\Delta}(r) = \log\left(k_\Delta r^{\frac{\Delta}{1-\Delta}} + \mathbb{1}\{q_\Delta > 0\} r^{\frac{1-q_\Delta}{q_\Delta}} \right),$$

where

$$k_\Delta := \left\lfloor \frac{1}{1-\Delta} \right\rfloor, \qquad q_\Delta := 1 - k_\Delta(1-\Delta).$$

The corresponding statistic in our notation is

$$S_{\mathrm{Li},\Delta} := \sum_{t\in\mathcal{T}} h^\star_{\mathrm{gum},\Delta}(r_t(w_t)).$$

This score is the log-likelihood ratio associated with the least-favorable distribution

$$P^\star_\Delta = (\underbrace{1-\Delta,\dots,1-\Delta}_{k_\Delta \text{ times}}, q_\Delta, 0, \dots, 0),$$

with the q_Δ term omitted when $q_\Delta = 0$. Therefore, $h^\star_{\mathrm{gum},\Delta}$ is not the same as Aaronson's original score h_{A}. Indeed,

$$h_{\mathrm{A}}(r) \to \infty \qquad \text{as } r \uparrow 1,$$

whereas

$$h^\star_{\mathrm{gum},\Delta}(r) \to \log\left(k_\Delta + \mathbb{1}\{q_\Delta > 0\}\right) \qquad \text{as } r \uparrow 1.$$

Thus, Li et al.'s score and Aaronson's original score are based on the same Gumbel pivot $r_t(w_t)$, but they lead to different cumulative detection statistics.

E.4 WATERMARK SCORE BY (LATTIMORE, 2026)

Lattimore (2026) proposed a refined detector for the Gumbel watermark based on a truncated power-law score. In our notation, the same Gumbel pivot is

$$r_t(w_t), \qquad t \in \mathcal{T}.$$

Instead of Aaronson's original per-token score

$$h_{\mathrm{A}}(r) := -\log(1-r),$$

Lattimore (Lattimore, 2026) considers

$$h_{\mathrm{PL},\epsilon}(r) = \min\left\{\epsilon^{-1/2}, (1-r)^{-1/2}\right\} - (2-\sqrt{\epsilon}),$$

and the corresponding statistic is

$$S_{\mathrm{PL},\epsilon} := \sum_{t\in\mathcal{T}} h_{\mathrm{PL},\epsilon}(r_t(w_t)).$$

Here the centering term $2-\sqrt{\epsilon}$ is chosen so that, under the null hypothesis,

$$\mathbf{E}[h_{\mathrm{PL},\epsilon}(U)] = 0, \qquad U \sim \mathrm{Unif}[0,1].$$

Thus, under the null and assuming independence across the scored positions,

$$\mathbf{E}[S_{\mathrm{PL},\epsilon}] = 0.$$

Both $S_{\mathrm{PL},\epsilon}$ and Aaronson's statistic

$$S_{\mathrm{A}} = \sum_{t\in\mathcal{T}} h_{\mathrm{A}}(r_t(w_t))$$

are model-agnostic statistics based on the same Gumbel pivot $r_t(w_t)$. However, they are not equivalent. In particular,

$$h_{\mathrm{A}}(r) \to \infty \qquad \text{as } r \uparrow 1,$$

whereas

$$h_{\mathrm{PL},\epsilon}(r) \le \epsilon^{-1/2} - (2-\sqrt{\epsilon})$$

is bounded because of the truncation. Therefore, Lattimore's truncated power-law score and Aaronson's original score use the same keyed randomness, but lead to different cumulative detection statistics.

F ALGORITHM IMPLEMENTATION

Algorithm 8: Compute Aaronson score, exact null p-value, and ANLPPT for keyed PFR watermark

Input : output sequence $w_{1:N}$, prompt length n_0, labeler Λ, secret key k, reference measure μ
Output : Aaronson score S_{A}, p-value p_{A}, ANLPPT $\mathrm{ANLPPT}_{\mathrm{A}}$ and $M = |\mathcal{T}|$

$S_{\mathrm{A}} \leftarrow 0$
$M \leftarrow 0$
for $t = n_0 + 1$ **to** N **do**
  $c_t \leftarrow w_{:t-1}$
  $\ell_t \leftarrow \Lambda(c_t)$
  $\Pi_t \leftarrow G(\mathsf{k}, \ell_t)$
  // recover keyed uniform value for the observed token w_t
  $\tau_t(w_t) \leftarrow \textsc{FirstArrival}(w_t, \Pi_t, \mu)$
  $r_t \leftarrow \exp(-\mu(w_t)\tau_t(w_t))$
  $S_{\mathrm{A}} \leftarrow S_{\mathrm{A}} - \log(1-r_t)$
  $M \leftarrow M + 1$
end
// Under H_0, $S_{\mathrm{A}} \sim \mathrm{Gamma}(M, 1)$
$p_{\mathrm{A}} \leftarrow \Gamma(M, S_{\mathrm{A}})/\Gamma(M)$
$\mathrm{ANLPPT}_{\mathrm{A}} \leftarrow -\frac{1}{M}\log p_{\mathrm{A}}$

Algorithm 9: Watermarkable Speculative Sampling via Poisson Processes

Input : lookahead K, output length N, target model family P, draft model family Q, initial prefix $w_{1:n}$, common reference measure μ on $\mathcal{Z}$ such that $P(\cdot|c), Q(\cdot|c) \ll \mu$ for all contexts c, labeler Λ, secret key k

while $n < N$ **do**

  $h \leftarrow w_{1:n}$ `// freeze block-start prefix`

  $L \leftarrow \min\{K,\, N - n\}$

  for $s = 1$ **to** L `// draft` L `tokens from shared PFR sources` **do**

    $c_s \leftarrow h \,\|\, \tilde{w}_{1:s-1}$

    $\ell_s \leftarrow \Lambda(c_s)$ `// includes repeated-context masking if used`

    $\Pi_s \leftarrow G(\mathsf{k}, \ell_s)$

    $\tilde{w}_s \leftarrow \mathrm{PFR}(Q(\cdot|c_s), \Pi_s, \mu)$

  end

  if $n + L < N$ **:**

    $c_{L+1} \leftarrow h \,\|\, \tilde{w}_{1:L}$

  end

  `// Prepare target-side evaluations for` $c_1, \ldots, c_L$ `and, if needed,` c_{L+1} `in parallel`

  $accepted \leftarrow$ **true**

  `// Verify each draft token using the same shared Poisson source`

  for $s = 1$ **to** L **do**

    $y_s \leftarrow \mathrm{PFR}(P(\cdot|c_s), \Pi_s, \mu)$

    $w_{n+1} \leftarrow y_s$; $n \leftarrow n + 1$ `// emit the target PFR winner`

    if $y_s \neq \tilde{w}_s$ **:**

      $accepted \leftarrow$ **false**

      break `// correction: later draft contexts are invalid`

    end

  end

  if $accepted$ ***and*** $n < N$ **:**

    $\ell_{L+1} \leftarrow \Lambda(c_{L+1})$

    $\Pi_{L+1} \leftarrow G(\mathsf{k}, \ell_{L+1})$

    $w_{n+1} \leftarrow \mathrm{PFR}(P(\cdot|c_{L+1}), \Pi_{L+1}, \mu)$

    $n \leftarrow n + 1$ `// bonus token`

  end

end

Algorithm 10: Watermarkable Multi-Draft Speculative Sampling (Formal)

Given : lookahead L, number of drafts B, output length N, target model P, draft model Q, initial prompt $w_{1:n}$, keyed Poisson source generator G, and secret key k

while $n < N$ **do**
  $h \leftarrow w_{1:n}$; $\ell \leftarrow \min\{L,\, N-n\}$
  $\mathcal{C}_0 \leftarrow \{h\}$; $\nu(h) \leftarrow B$ `// root context; all` B `homogeneous drafts start here`
  for $s = 1$ **to** ℓ **do**
    $\mathcal{C}_s \leftarrow \emptyset$
    for $c \in \mathcal{C}_{s-1}$ **do**
      $\Pi(c) \leftarrow G(\mathsf{k}, c)$ `// context-indexed keyed source shared by draft and target at context` c
      for $i = 1$ **to** $\nu(c)$ **do**
        $\tilde{w}(c,i) \leftarrow \mathrm{MPFR}(Q(\cdot|c), \Pi(c), i)$
      $\mathcal{D}(c) \leftarrow \{\tilde{w}(c,1), \ldots, \tilde{w}(c,\nu(c))\}$ `// distinct next-token proposals from context` c
      for $u \in \mathcal{D}(c)$ **do**
        $\nu(c \,\|\, u) \leftarrow \big|\{\, i \in \{1,\ldots,\nu(c)\} : \tilde{w}(c,i) = u \,\}\big|$
        `// how many drafts move to child context` $c\|u$
        $\mathcal{C}_s \leftarrow \mathcal{C}_s \cup \{\, c \,\|\, u \,\}$
  for $c \in \mathcal{C}_0 \cup \cdots \cup \mathcal{C}_{\ell-1}$ **do** `// compute target-side winner at every internal context`
    $y(c) \leftarrow \mathrm{MPFR}(P(\cdot|c), \Pi(c), 1)$
    `// target-side mapped-Poisson winner at context` c
  $c^\star \leftarrow h$; *accepted* $\leftarrow$ **true**
  `// start following the unique realized target path from the root`
  for $s = 1$ **to** ℓ **do**
    $a_s \leftarrow \mathbb{1}\{y(c^\star) \in \mathcal{D}(c^\star)\}$
    `// accept if the target winner is among the drafts at the current context`
    $w_{n+1} \leftarrow y(c^\star)$; $n \leftarrow n+1$
    `// emit the target-side token, not a draft token`
    if $a_s = 0$ **:**
      accepted $\leftarrow$ **false** and **break** `// first rejection ends the speculative block`
    if $n = N$ **:**
      break
    $c^\star \leftarrow c^\star \,\|\, y(c^\star)$ `// move to the next realized context`
  if *accepted* ***and*** $n < N$ **:** `// if all` ℓ `steps were accepted, emit one extra target token`
    $\Pi^\star \leftarrow G(\mathsf{k}, c^\star)$
    $w_{n+1} \leftarrow \mathrm{MPFR}(P(\cdot|c^\star), \Pi^\star, 1)$
    $n \leftarrow n+1$ `// bonus token`